

**RESEARCH PROJECT
REPORT ON
"PREDICTIVE ANALYSIS OF CUSTOMER CONVERSION IN THE
INSURANCE INDUSTRY"**

SUBMITTED TO



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SUBMITTED BY

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DECLARATION

I, the undersigned, hereby declare that the Project Report entitled “**PREDICTIVE ANALYSIS OF CUSTOMER CONVERSION IN INSURANCE INDUSTRY**” written and submitted by me to the Savitribai Phule Pune University, in partial fulfilment of the requirement for the award of the degree of Master of Business Administration, is my original work and the conclusions drawn therein are based on the material collected by myself.

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EXECUTIVE SUMMARY

This study, titled “**Predictive Analysis of Customer Conversion in the Insurance Industry**” focuses on understanding the major factors that influence whether potential insurance leads convert into paying customers. In today’s highly competitive insurance market, companies invest significant resources in lead generation, digital marketing, and customer acquisition activities. However, many organizations still face difficulties in identifying which customers are most likely to convert and which factors play the strongest role in influencing purchase decisions. This research aims to address this challenge through the use of predictive analytics and secondary data analysis.

The main problem identified in this study is that insurance firms often possess large volumes of customer and lead-related data, but they are unable to fully utilize this information for effective decision-making. Variables such as premium amount, discounts, customer engagement, credit score, policy type, and lead source interact in complex ways, making it difficult for traditional analysis methods to accurately predict conversion behavior. As a result, businesses may allocate marketing resources inefficiently, lose potential customers, and struggle to optimize pricing and promotional strategies.

The purpose of this study is to examine how demographic, financial, behavioral, and policy-related variables affect customer conversion in the insurance sector. The research uses secondary data consisting of 10,000 insurance customer records with 27 variables and no missing values. The study adopts a descriptive and analytical research design. Data analysis was performed using Python-based analytical techniques, including descriptive statistics, correlation analysis, hypothesis testing, visualization, and machine learning classification models such as Logistic Regression and Random Forest.

The findings of the study reveal several important insights regarding customer conversion behavior. Premium amount was identified as one of the most influential factors affecting conversion. Higher premium amounts significantly reduced the likelihood of customer conversion, indicating that customers are highly price-sensitive when selecting insurance products. Discounts were also found to positively influence conversion rates, suggesting that targeted discount strategies can improve customer acquisition performance.

Customer engagement variables showed mixed results. Website visits had a statistically significant positive relationship with conversion, demonstrating that digitally engaged customers are more likely to purchase insurance policies. However, variables such as inquiries and quotes requested did not significantly affect conversion outcomes. Policy type emerged as the most significant categorical variable, with Liability-Only policies showing higher conversion rates compared to Full Coverage policies. Regional differences and lead sources showed some variation in conversion rates, although these differences were not statistically significant at the conventional level.

The machine learning models used in the study achieved moderate predictive performance. Logistic Regression produced slightly better results than Random Forest, though both models indicated that customer conversion behavior is influenced by multiple complex factors beyond

those captured in the dataset. The findings suggest that additional behavioral and psychological variables may further improve predictive accuracy in future studies.

The significance of this study lies in both its practical and academic contributions. From a business perspective, the research provides valuable insights that can help insurance companies improve customer targeting, pricing strategies, discount planning, and digital engagement practices. The study supports better decision-making by identifying the variables that most strongly influence conversion probability. Organizations can use these insights to optimize marketing investments, reduce customer acquisition costs, and improve overall conversion performance.

From an academic perspective, the study demonstrates how predictive analytics and secondary data analysis can be effectively applied in the insurance industry to understand customer behavior. It contributes to the growing body of research on business analytics, customer conversion prediction, and data-driven decision-making in financial services.

Overall, this project highlights the growing importance of predictive analytics in the insurance sector and emphasizes the need for organizations to adopt data-driven approaches for improving customer acquisition and competitive advantage. The study provides meaningful recommendations for insurers, marketers, and business analysts seeking to better understand and predict customer conversion behavior in a rapidly evolving digital marketplace.

CHAPTER 1 – INTRODUCTION

1.1 Background of the Study

In the modern business environment, the insurance industry has become increasingly competitive, technology-driven, and customer-focused. Insurance companies today operate in a rapidly evolving digital economy where customer expectations, market competition, and technological advancements continue to influence business performance. Customers now expect personalized insurance products, affordable premium pricing, faster services, transparent communication, and seamless digital experiences. As a result, insurance organizations are increasingly shifting from traditional decision-making methods toward data-driven analytical approaches that improve customer acquisition, operational efficiency, and business growth.

The growth of digital technologies such as online insurance platforms, mobile applications, customer relationship management systems, artificial intelligence, and digital marketing has enabled insurance companies to generate large volumes of customer-related data. This data includes customer demographics, website activity, policy preferences, credit information, premium details, discounts offered, customer inquiries, engagement patterns, and transaction records. Such information provides businesses with significant opportunities to better understand customer behavior and improve strategic decision-making processes.

One of the major challenges faced by insurance companies today is customer conversion. Customer conversion refers to the process through which a prospective lead successfully becomes a paying customer by purchasing an insurance policy. Insurance organizations invest substantial financial resources in lead generation campaigns, advertisements, online promotions, customer engagement activities, and sales operations. However, not all leads generated by companies result in successful conversions. Many customers discontinue the buying process due to factors such as high premium costs, unsuitable policy types, low customer trust, insufficient engagement, lack of personalized offers, or better alternatives provided by competitors.

Traditional business decision-making methods often rely on managerial assumptions, experience, or basic reporting systems, which may not be sufficient to analyze the complexity of modern customer behavior. Therefore, organizations are increasingly adopting predictive analytics and machine learning techniques to identify patterns in customer data and improve business outcomes. Predictive analytics enables businesses to analyze historical customer information, identify important variables affecting customer behavior, forecast future outcomes, and support evidence-based managerial decisions.

In the insurance sector, predictive analytics helps organizations determine which customers are more likely to purchase policies, which marketing strategies are most effective, and how engagement behavior influences customer conversion outcomes. By using analytical models, insurance companies can improve customer targeting, optimize pricing strategies, personalize promotional offers, reduce acquisition costs, and enhance customer relationship management practices. Organizations that effectively use analytics gain a strong competitive advantage in an increasingly data-driven business environment.

The present study focuses on “Predictive Analysis of Customer Conversion in the Insurance Industry.” The research examines how demographic, behavioral, financial, and policy-related variables influence customer conversion probability. Variables such as credit score, total discounts, policy type, lead source, premium amount, website visits, inquiries, quotes requested, marital status, and region are analyzed to understand their relationship with customer conversion behavior. The study applies secondary data analytics, statistical analysis, and machine learning techniques to generate insights that can support effective business decision-making within the insurance industry.

1.2 Meaning of Predictive Analytics in Customer Conversion

Predictive analytics refers to the process of using historical data, statistical techniques, machine learning algorithms, and analytical tools to identify patterns and predict future outcomes. In modern business environments, predictive analytics helps organizations forecast customer behavior, operational performance, market trends, and business risks using available data.

In the insurance industry, predictive analytics is widely used for fraud detection, pricing optimization, customer acquisition, claims management, customer retention, and risk assessment. One of the most important applications of predictive analytics is predicting customer conversion behavior. Insurance companies interact with thousands of potential customers through online platforms, websites, digital campaigns, agents, referrals, and customer inquiries. Predictive analytics helps organizations identify which customers are most likely to convert into policyholders.

Traditional customer conversion decisions were often based on assumptions, intuition, or manual evaluation methods. However, modern business competition requires more accurate, faster, and evidence-based decision-making approaches. Predictive analytics enables organizations to analyze multiple customer-related variables simultaneously and identify the factors that significantly influence customer conversion outcomes.

Predictive analytics can help answer important business questions such as:

- Which customers are most likely to purchase insurance policies?
- How does credit score influence customer conversion probability?
- Do higher discounts increase conversion rates?
- Which lead source generates better conversion outcomes?
- How does customer engagement affect purchase behavior?
- Which policy type has the highest conversion rate?
- How do demographic factors influence customer decisions?
- What customer characteristics are associated with successful conversion?

Instead of depending only on assumptions or traditional marketing approaches, organizations can use predictive models to make smarter and more profitable business decisions.

Core Definition

Predictive Analytics can be defined as:

“A data-driven analytical approach that uses statistical models, machine learning techniques, and historical information to predict future outcomes and support business decision making.”

This means organizations use customer data and analytical tools to improve forecasting accuracy, reduce uncertainty, and optimize business strategies.

1.3 Why the Topic is Important?

The topic “Predictive Analysis of Customer Conversion in the Insurance Industry” is highly important in today’s business environment because customer acquisition and retention directly influence organizational growth, profitability, and long-term sustainability. The insurance industry has become increasingly competitive due to the expansion of digital platforms, online insurance providers, and changing customer expectations. Companies now require more efficient methods to identify high-potential customers and improve conversion performance.

One major reason this topic is important is the rapid growth in customer data availability. Insurance companies collect large amounts of customer information through websites, mobile applications, online interactions, digital advertisements, customer service platforms, and policy transactions. However, many organizations struggle to transform raw data into meaningful business insights. Predictive analytics enables businesses to convert data into actionable information that supports better managerial decisions.

Another important reason is changing customer behavior. Modern customers compare multiple insurance providers before making purchasing decisions. They expect affordable premium pricing, attractive discounts, personalized recommendations, faster responses, and convenient online services. Customers also demonstrate different engagement levels based on website visits, inquiries, quotes requested, and communication quality. Understanding these behavioral patterns is essential for improving customer conversion strategies.

The topic is also important because predictive analytics improves operational efficiency and marketing effectiveness. Insurance organizations spend significant amounts on advertising campaigns, lead generation, and customer acquisition activities. Without proper analytical systems, businesses may waste resources targeting low-potential customers. Predictive analytics enables organizations to prioritize customers who are more likely to convert and optimize marketing investments.

From a strategic perspective, predictive analytics provides competitive advantage. Organizations that effectively use analytical models can improve customer targeting, personalize offers, optimize pricing decisions, and strengthen customer relationships. Companies that fail to adopt data-driven strategies may lose market share to more technologically advanced competitors.

Furthermore, the insurance industry is undergoing rapid digital transformation. Technologies such as artificial intelligence, machine learning, automation, cloud computing, and big data analytics are reshaping business operations within the industry. As a result, understanding predictive analytics has become increasingly important for managers, analysts, marketers, and researchers.

1.4 Overview of Predictive Analytics in the Insurance Industry

The insurance industry has increasingly adopted predictive analytics to improve operational efficiency, customer satisfaction, business performance, and strategic decision-making. Insurance companies today use analytical models across multiple business functions, including underwriting, fraud detection, pricing optimization, customer retention, claims management, and customer conversion prediction.

One of the major applications of predictive analytics is customer conversion analysis. Insurance organizations generate leads through multiple channels such as online platforms, digital advertisements, websites, referrals, agents, and customer inquiries. Predictive analytics helps businesses evaluate these leads and identify customers who are more likely to purchase insurance policies.

Insurance companies use customer-related variables such as:

- Credit score
- Total discounts
- Customer age
- Marital status
- Website visits
- Inquiries
- Quotes requested
- Policy type
- Region
- Source of lead
- Premium amount
- Claims frequency

These variables help organizations better understand customer behavior and predict conversion probability more accurately.

Machine learning techniques such as Logistic Regression, Decision Trees, Random Forest, and Classification Models are widely used in insurance analytics. These techniques analyze historical customer data and identify patterns that may not be visible through traditional analysis methods.

Predictive analytics also supports pricing optimization within the insurance industry. Organizations can determine how premium pricing and discount structures influence customer

purchase decisions. Similarly, customer engagement analytics helps companies understand how digital interactions and website behavior affect customer conversion outcomes.

As insurance companies continue to digitize operations and customer services, predictive analytics is becoming an essential tool for achieving customer-centric growth, improving operational efficiency, and maintaining long-term competitive advantage.

1.5 Why Customer Conversion Analysis is Important

Customer conversion analysis is important because converting potential leads into paying customers directly affects organizational profitability, business growth, and market sustainability. Insurance companies invest large amounts of money in digital marketing campaigns, advertisements, sales teams, promotional offers, and lead generation activities. However, if organizations fail to convert potential customers successfully, these investments may not generate expected business returns.

Conversion analysis helps businesses identify the major factors influencing customer purchase decisions. It enables organizations to recognize barriers that prevent conversion and opportunities that improve customer acquisition performance. Businesses can then develop targeted strategies to improve pricing, engagement, communication, and service delivery.

Another important reason customer conversion analysis is significant is the increasing cost of customer acquisition in highly competitive digital markets. Insurance organizations must therefore maximize the effectiveness of every lead generated. Predictive analytics enables companies to prioritize customers with higher conversion probability and reduce unnecessary marketing expenditure.

Customer conversion analysis also improves personalization strategies. Different customers respond differently to discounts, premium pricing, engagement methods, policy types, and communication channels. By analyzing customer behavior patterns, organizations can provide more customized offers and improve customer satisfaction.

In addition, customer conversion analysis supports long-term business planning. It helps organizations forecast sales performance, evaluate marketing effectiveness, and design better customer acquisition and retention strategies. Companies that successfully improve conversion rates achieve stronger market positioning and higher profitability.

Overall, customer conversion analysis has become an essential component of modern insurance business strategy because it supports smarter, faster, and more data-driven decision-making processes.

1.6 Scope of the Study

The scope of the present study defines the boundaries and coverage of the research. This study focuses on predictive analysis of customer conversion in the insurance industry using

secondary data analytics. The research examines how demographic, behavioral, financial, and policy-related variables influence the probability of customer conversion.

The study is based on a secondary dataset consisting of 10,000 customer records related to insurance conversion behavior. The dataset contains multiple variables associated with customer demographics, engagement behavior, risk profile, and policy-related characteristics.

The study includes variables such as:

- Age
- Is_Senior
- Marital_Status
- Region
- Source_of_Lead
- Time_Since_First_Contact
- Website_Visits
- Inquiries
- Quotes_Requested
- Claims_Frequency
- Claims_Severity
- Prior_Insurance
- Credit_Score
- Policy_Type
- Premium_Amount
- Total_Discounts
- Conversion_Status

The research mainly focuses on identifying patterns and relationships that influence whether a customer converts into a paying policyholder. Statistical techniques such as descriptive analysis, correlation analysis, hypothesis testing, and machine learning classification models are applied to analyze the dataset.

The study specifically tests hypotheses related to the impact of credit score, total discounts, policy type, and lead source on customer conversion probability. Binary Logistic Regression is used as the primary analytical technique to evaluate these relationships.

The research is limited to secondary data analysis and does not include primary data collection methods such as surveys or interviews. The findings are entirely based on the available dataset and analytical methods used within the study. The study also focuses specifically on customer conversion behavior within the insurance sector and may not fully represent customer behavior in other industries.

Another important scope area is the application of predictive analytics and machine learning techniques. Models such as Logistic Regression and Random Forest are used to evaluate predictive performance and identify major conversion drivers. The study aims to provide practical insights that can help insurance companies improve customer acquisition strategies, pricing decisions, and customer engagement practices.

Overall, the scope of the study includes an analytical examination of customer conversion behavior using predictive analytics within the insurance industry.

1.7 Need for the Study

The need for this study arises from the growing importance of customer acquisition and predictive analytics within the insurance industry. Modern insurance companies operate in a highly competitive environment where attracting and retaining customers has become increasingly difficult. Organizations generate large numbers of leads through online platforms, websites, agents, referrals, and digital marketing campaigns, but many of these leads fail to convert into paying customers.

One major reason for conducting this study is that many insurance companies possess large volumes of customer data but struggle to use it effectively for business decision-making. Businesses often collect information related to customer demographics, engagement activities, credit scores, discounts, premium preferences, and digital interactions, yet fail to convert this information into actionable insights. Predictive analytics helps solve this problem by identifying patterns and predicting customer behavior more accurately.

Another need for this study is the increasing importance of digital transformation within the insurance industry. Customers today expect personalized recommendations, affordable premium pricing, attractive discounts, quick responses, and convenient digital experiences. Companies that fail to understand customer expectations may lose potential customers to competitors. Therefore, analyzing customer conversion behavior has become essential for improving customer engagement and business performance.

The study is also needed because customer acquisition costs are increasing rapidly. Insurance companies invest substantial amounts in advertisements, promotions, and customer acquisition campaigns. Without proper analytical systems, organizations may waste resources targeting low-potential customers. Predictive analytics enables businesses to focus on high-probability leads and improve marketing efficiency.

In addition, there is a growing need for academic research that combines predictive analytics, business analytics, and customer behavior analysis within the insurance sector.

This study contributes to existing knowledge by applying statistical analysis and machine learning techniques to understand customer conversion patterns using secondary data.

CHAPTER 2 – LITERATURE REVIEW

2.1 Introduction

Predictive analytics and customer conversion analysis have become important areas of study in modern business management because organizations increasingly rely on data, analytics, and machine learning techniques to improve business decisions. In today's digital economy, insurance companies generate large volumes of customer-related information through websites, online platforms, mobile applications, customer relationship management systems, digital advertisements, and lead generation campaigns. However, merely collecting customer data does not guarantee improved business performance. Competitive advantage depends on how effectively organizations convert available information into meaningful business insights and customer acquisition strategies.

In the insurance industry, customer conversion is considered one of the most important indicators of business growth and profitability. Insurance organizations invest heavily in customer acquisition campaigns, digital marketing, promotional offers, and customer engagement activities. However, many prospective customers fail to convert into paying policyholders due to factors such as premium pricing, low engagement, policy preferences, weak communication, or competitive alternatives.

Researchers have increasingly emphasized the importance of predictive analytics, machine learning, customer behavior analysis, and data-driven decision-making in improving customer conversion outcomes. Predictive analytics helps organizations identify high-potential customers, optimize pricing strategies, personalize offers, improve engagement, and reduce customer acquisition costs. Machine learning models such as Logistic Regression, Decision Trees, and Random Forest are widely used to analyze customer-related variables and predict conversion probability.

The present literature review examines previous studies related to predictive analytics, customer conversion behavior, machine learning applications, customer engagement, pricing strategies, insurance analytics, and data-driven decision-making within the insurance industry.

2.2 Concept of Predictive Analytics

Predictive analytics refers to the use of historical data, statistical techniques, machine learning algorithms, and analytical tools to forecast future outcomes and support managerial decision-making. Davenport and Harris (2007) stated that organizations using analytical decision-making approaches achieve better operational efficiency and competitive advantage compared to firms relying only on traditional judgment methods.

Brynjolfsson, Hitt, and Kim (2011) found that data-driven organizations demonstrate higher productivity and improved business performance. Provost and Fawcett (2013) emphasized that competitive advantage does not come from owning data alone, but from effectively interpreting patterns and generating actionable insights.

Rousseau (2006) linked predictive analytics with evidence-based management, where organizations use factual information rather than assumptions while making business decisions. Sharma, Mithas, and Kankanhalli (2014) further argued that analytical culture, employee capability, and leadership support are important for successful implementation of predictive analytics systems.

Researchers also suggest that predictive analytics improves forecasting accuracy, operational efficiency, customer targeting, and strategic planning across industries.

2.3 Predictive Analytics and Organizational Performance

Predictive analytics has significantly influenced organizational decision-making and business performance. Chen, Chiang, and Storey (2012) explained that business intelligence and analytics systems help transform operational data into actionable knowledge that supports managerial decisions.

McAfee and Brynjolfsson (2012) argued that forecasting, optimization, and experimentation become more effective when organizations use large datasets and predictive models. Akter et al. (2016) found that analytics capability positively influences customer integration, operational performance, and financial outcomes.

Wamba et al. (2017) concluded that big data analytics improves innovation capability and organizational agility. Mikalef et al. (2019) reported that strategic alignment between analytics resources and business goals creates stronger competitive advantage and better organizational performance.

Studies indicate that predictive analytics enables organizations to improve decision-making quality, reduce uncertainty, and enhance long-term business growth.

2.4 Customer Conversion Analysis

Customer conversion analysis refers to the process of examining factors that influence whether potential customers successfully convert into paying customers. Researchers have emphasized that customer conversion is directly related to organizational profitability and customer acquisition efficiency.

Kumar et al. (2013) stated that predictive customer models help businesses allocate resources more effectively and improve customer acquisition strategies. Lemon and Verhoef (2016) found that customer journey analysis improves engagement and satisfaction across multiple customer touchpoints.

Rust and Huang (2014) argued that organizations can improve customer lifetime value through personalization and targeted engagement strategies. Wedel and Kannan (2016) reported that data-driven personalization significantly improves customer engagement in digital markets.

Studies suggest that customer conversion analysis helps organizations identify high-potential customers, optimize marketing investments, and improve customer acquisition performance.

2.5 Machine Learning in Business Analytics

Machine learning has become one of the most important components of predictive analytics and business intelligence systems. Researchers have increasingly applied machine learning techniques to customer behavior analysis, fraud detection, recommendation systems, and predictive modeling.

James et al. (2013) explained that machine learning models such as Logistic Regression, Decision Trees, and Random Forest improve prediction accuracy in business analytics applications. Han, Kamber, and Pei (2012) stated that classification techniques help organizations identify hidden patterns within customer datasets.

Aggarwal (2015) argued that machine learning improves customer segmentation, predictive analysis, and decision support systems. Kotu and Deshpande (2018) found that predictive modeling techniques significantly improve customer conversion prediction and marketing effectiveness.

Researchers also note that machine learning techniques help organizations automate decision-making processes and improve forecasting accuracy.

2.6 Customer Engagement and Digital Behavior

Customer engagement plays an important role in influencing customer conversion behavior within digital platforms. Researchers suggest that customer interaction with websites, mobile applications, digital advertisements, and communication systems significantly affects purchase decisions.

Lemon and Verhoef (2016) reported that customer engagement across multiple digital touchpoints improves customer satisfaction and conversion outcomes. Moe and Fader (2004) found that website browsing behavior and online interaction patterns provide important insights into customer purchase intentions.

Court et al. (2009) argued that digital engagement influences customer decision journeys and brand relationships. Verhoef, Reinartz, and Krafft (2010) emphasized that customer relationship management systems improve engagement and customer retention.

Studies indicate that variables such as website visits, inquiries, quotes requested, and online interactions are important predictors of customer conversion probability.

2.7 Insurance Analytics

The insurance industry has increasingly adopted predictive analytics to improve pricing, fraud detection, customer acquisition, claims management, and customer retention. Researchers have

emphasized that insurance companies generate large volumes of customer-related data that can support strategic decision-making.

Davenport (2014) stated that analytics-driven insurance companies achieve better operational efficiency and customer targeting. Berson, Smith, and Thearling (2000) found that data mining techniques improve customer segmentation and risk assessment within insurance markets.

Ngai et al. (2009) reported that predictive analytics helps insurance companies improve customer acquisition and fraud detection performance. Shmueli, Bruce, and Patel (2016) explained that predictive models support customer conversion analysis and pricing optimization within the insurance sector.

Studies suggest that insurance analytics improves operational efficiency, customer experience, and profitability.

2.8 Credit Score and Customer Conversion

Credit score is considered an important financial indicator influencing customer conversion behavior in the insurance industry. Researchers suggest that customers with stronger credit profiles are more likely to purchase insurance policies and maintain long-term relationships with organizations.

Anderson (2007) found that credit-related variables significantly influence customer purchase decisions and financial behavior. Avery, Calem, and Canner (2004) reported that higher credit scores are associated with lower financial risk and improved customer stability.

Crook, Edelman, and Thomas (2007) argued that credit scoring systems improve predictive accuracy in customer analysis and financial decision-making. Hand and Henley (1997) explained that credit scoring models are widely used for predicting customer behavior and financial reliability.

Studies indicate that higher credit scores positively influence customer conversion probability and customer acquisition outcomes.

2.9 Discounts and Pricing Strategies

Pricing and discount strategies significantly influence customer purchase decisions and conversion behavior. Researchers have emphasized that discounts improve perceived value and customer motivation.

Kotler and Keller (2016) stated that pricing strategies directly affect customer behavior and market competitiveness. Monroe (2003) argued that customers evaluate value based on pricing fairness and promotional offers.

Blattberg and Neslin (1990) found that sales promotions and discounts positively influence customer acquisition and short-term purchase behavior. Gupta and Cooper (1992) reported that effective discount strategies improve customer response rates and conversion outcomes.

Studies suggest that higher discounts increase customer purchase probability and improve conversion performance.

2.10 Policy Type and Customer Preferences

Insurance policy type is another important factor influencing customer conversion behavior. Researchers suggest that customer preferences differ based on coverage level, pricing structure, and perceived value.

Zeithaml (1988) argued that customers evaluate products based on perceived quality and benefits. Schiffman and Kanuk (2007) explained that customer purchase decisions are influenced by product characteristics and perceived utility.

Research within insurance markets indicates that customers often prefer policies offering broader coverage and better financial protection. Full Coverage policies may therefore generate higher conversion rates compared to Liability-Only policies due to higher perceived value and security.

Studies suggest that policy-related variables significantly influence customer acquisition and customer satisfaction outcomes.

2.11 Lead Source and Customer Acquisition

Lead generation source plays a critical role in customer acquisition and conversion analysis. Insurance companies generate leads through agents, websites, digital advertisements, referrals, and online platforms.

Chaffey and Ellis-Chadwick (2019) stated that digital marketing channels improve customer targeting and lead generation effectiveness. Ryan and Jones (2016) argued that online lead generation systems provide better customer insights and engagement opportunities.

Researchers suggest that Online leads often demonstrate higher engagement levels due to direct interaction with digital platforms. Agent-generated leads may involve personal communication and relationship-based selling approaches.

Studies indicate that lead source significantly influences customer conversion probability and customer acquisition performance.

2.12 Customer Demographics and Conversion Behavior

Customer demographics influence customer preferences, engagement patterns, and purchase decisions. Researchers have emphasized that variables such as age, marital status, region, and income significantly affect customer behavior.

Kotler and Armstrong (2018) reported that demographic segmentation improves customer targeting and marketing effectiveness. Solomon (2017) explained that customer behavior

varies across demographic groups due to differences in lifestyle, preferences, and financial priorities.

Research suggests that demographic variables influence insurance purchasing decisions and customer conversion outcomes. Senior customers, married individuals, and customers from different regions may demonstrate different insurance preferences and engagement behavior.

Studies indicate that demographic analysis improves customer segmentation and predictive modeling accuracy.

2.13 Customer Relationship Management and Conversion

Customer Relationship Management (CRM) systems help organizations improve customer engagement, communication, and conversion performance. Researchers suggest that CRM systems support customer acquisition and retention strategies.

Payne and Frow (2005) argued that CRM improves customer relationship quality and organizational performance. Buttle and Maklan (2015) explained that CRM systems enable organizations to analyze customer behavior and personalize interactions.

Reinartz, Krafft, and Hoyer (2004) found that customer-focused relationship management positively influences customer acquisition and retention outcomes. Studies suggest that CRM systems improve customer communication, engagement, and conversion probability.

2.14 Research Gap

Although significant literature exists on predictive analytics, customer behavior analysis, machine learning, and business intelligence, several important research gaps remain within the insurance industry.

Most previous studies examine predictive analytics, pricing strategies, customer engagement, or customer behavior separately rather than integrating demographic, behavioral, financial, and policy-related variables into a unified customer conversion framework.

Another major research gap is the limited application of machine learning and predictive analytics specifically for customer conversion analysis within the insurance sector. Existing studies often focus on fraud detection, claims management, and risk assessment instead of customer acquisition and conversion prediction.

Limited research is also available comparing the impact of variables such as credit score, discounts, policy type, and lead source on customer conversion probability. Few studies examine how these variables collectively influence customer purchase behavior within insurance markets.

Many previous studies are conceptual and descriptive in nature, while fewer studies apply measurable predictive models such as Logistic Regression and Random Forest for customer conversion analysis.

In addition, limited research exists regarding how insurance companies can use predictive analytics to improve customer acquisition strategies, optimize pricing decisions, and reduce customer acquisition costs in digital environments.

Therefore, the present study attempts to bridge these gaps by providing an evidence-based and analytical examination of customer conversion behavior using predictive analytics and machine learning techniques within the insurance industry.

CHAPTER 3-OBJECTIVES, HYPOTHESES AND CONCEPTUAL FRAMEWORK

3.1 Introduction

After reviewing previous literature on predictive analytics, customer behavior analysis, insurance marketing, machine learning applications, and customer conversion models, it becomes clear that modern organizations increasingly depend on data as a strategic business resource. Insurance companies today generate large volumes of information through customer interactions, websites, digital platforms, mobile applications, online advertisements, lead generation systems, and customer relationship management platforms. However, merely collecting customer data does not guarantee improved business performance. Competitive advantage depends on how effectively organizations convert available information into timely and accurate business decisions.

In the insurance industry, customer conversion has become one of the most important indicators of organizational growth and profitability. Insurance companies invest significant resources in digital marketing campaigns, promotional activities, customer acquisition programs, online platforms, and sales operations. However, many potential customers fail to convert into paying policyholders due to factors such as premium pricing, low engagement, unsuitable policy types, weak communication, or competitive alternatives.

Modern insurance organizations increasingly use predictive analytics and machine learning techniques to understand customer behavior and improve customer conversion outcomes. Predictive analytics helps organizations identify high-potential customers, improve pricing strategies, optimize customer targeting, personalize offers, and enhance customer engagement. Machine learning techniques such as Logistic Regression, Decision Trees, and Random Forest models help analyze customer-related variables and predict conversion probability more accurately.

The present study focuses on “Predictive Analysis of Customer Conversion in the Insurance Industry.” The study examines how demographic, behavioral, financial, and policy-related variables influence customer conversion probability. Variables such as credit score, total discounts, policy type, lead source, website visits, inquiries, premium amount, and customer demographics are analyzed to understand their relationship with customer conversion behavior.

This chapter presents the problem statement, need for investigation, research objectives, research questions, hypotheses, variables, conceptual framework, and significance of the study.

3.2 Problem Statement

Despite the growing adoption of predictive analytics and digital technologies, many insurance companies still face difficulties in implementing effective data-driven customer conversion systems. Organizations often collect large amounts of customer and operational data but fail to convert this information into practical managerial decisions.

Insurance companies generate leads through websites, digital advertisements, online platforms, referrals, agents, and promotional campaigns. However, many organizations struggle to identify the factors that significantly influence customer conversion behavior. Variables such as credit score, discounts, customer engagement, policy type, premium amount, and lead source interact in complex ways that traditional decision-making approaches may fail to analyze accurately.

Insurance organizations increasingly rely on predictive analytics and machine learning techniques to improve customer acquisition and conversion performance. However, many companies still face challenges in identifying high-potential customers, optimizing pricing strategies, improving engagement methods, and allocating marketing resources effectively.

This research therefore aims to identify:

- How demographic, behavioral, and financial variables influence customer conversion probability
- Whether higher credit scores positively influence customer conversion
- Whether higher discounts improve customer conversion rates
- Whether Full Coverage policy type has higher conversion rates than Liability-Only policy type
- Whether Online leads convert more effectively than Agent leads
- How predictive analytics and machine learning techniques support customer conversion prediction and business decision-making

3.3 Need for the Problem Investigation

In the present competitive business environment, insurance companies generate huge volumes of customer-related data every day. However, many organizations are unable to use this information effectively while making business decisions. Weak use of customer data can result in poor marketing strategy, lower customer conversion rates, operational inefficiency, increased acquisition costs, and reduced profitability.

Insurance companies today operate in a highly competitive digital marketplace where customers compare multiple insurance providers before purchasing policies. Customers expect affordable pricing, attractive discounts, personalized recommendations, quick responses, and convenient digital experiences. Organizations that fail to understand customer behavior may lose potential customers to competitors.

Predictive analytics has become an important tool for improving customer conversion and business performance within the insurance industry. By analyzing variables such as customer

demographics, engagement behavior, credit score, discounts, policy type, and lead source, organizations can identify customers who are more likely to purchase insurance policies.

The need for this investigation is to understand how insurance companies can convert customer data into effective business decisions and improve customer conversion performance. The study can help identify important variables, analytical techniques, and predictive models that support customer acquisition and conversion management.

The findings may be useful for insurance managers, analysts, researchers, students, and organizations interested in adopting predictive analytics and machine learning techniques for improving customer conversion and business performance.

3.4 Research Objectives

The present study aims to achieve the following objectives:

1. To examine the role of predictive analytics in customer conversion analysis within the insurance industry.
2. To analyze the impact of credit score on customer conversion probability.
3. To evaluate the influence of total discounts on customer conversion behavior.
4. To identify the relationship between policy type and customer conversion rates.
5. To assess whether Online leads convert more effectively than Agent leads.
6. To examine the influence of demographic and engagement variables on customer conversion outcomes.
7. To apply machine learning and Logistic Regression techniques for predicting customer conversion behavior.
8. To provide data-driven recommendations for improving customer acquisition strategies in the insurance industry.

3.5 Research Questions

The study seeks to answer the following questions:

1. How does predictive analytics support customer conversion analysis in the insurance industry?
2. Does higher credit score positively influence customer conversion probability?
3. Do higher total discounts improve customer conversion rates?

4. Is there a significant relationship between policy type and customer conversion behavior?
5. Do Online leads convert more effectively than Agent leads?
6. How do demographic and engagement variables influence customer conversion outcomes?
7. How effective is Logistic Regression in predicting customer conversion behavior?

3.6 Research Hypotheses

Based on the research objectives and data analysis, the following hypotheses are proposed:

H₀: Higher credit score does not significantly influence customer conversion probability.

H₁: Higher credit score positively influences customer conversion probability.

H₂: Customers receiving higher total discounts are more likely to convert.

H₃: Full Coverage policy type has a higher conversion rate than Liability-Only policy type.

H₄: Online leads convert at a higher rate than Agent leads after controlling for engagement and demographic variables.

3.7 Variables of the Study

Variables are measurable factors considered to examine the relationship between predictive analytics variables and customer conversion outcomes.

A. Independent Variables

1. Credit Score – Represents the financial credibility of customers.
2. Total Discounts – Indicates the total discounts offered on insurance policies.
3. Policy Type – Represents the type of insurance policy selected by customers.
4. Source of Lead – Indicates the source through which customer leads are generated.
5. Customer Demographics – Includes age, marital status, senior citizen category, and region.
6. Engagement Behavior – Includes website visits, inquiries, quotes requested, and time since first contact.
7. Risk Profile – Includes claims frequency, claims severity, prior insurance, and related risk indicators.

8. Premium Amount – Represents the premium pricing associated with insurance policies.

B. Dependent Variables

1. Conversion Status – Indicates whether the customer purchased an insurance policy or not.

3.8 Conceptual Framework

The conceptual framework of the study explains the relationship between predictive analytics variables and customer conversion outcomes within the insurance industry.

It assumes that effective use of customer demographics, engagement behavior, financial variables, and policy-related information leads to improved customer conversion prediction and business decision-making. Variables such as credit score, discounts, lead source, policy type, premium amount, and customer engagement significantly influence customer purchase behavior.

Predictive analytics and machine learning techniques help organizations identify customer behavior patterns, evaluate conversion probability, and improve customer targeting strategies. Higher customer engagement, attractive discounts, strong credit profiles, and effective lead generation sources positively contribute to improved customer conversion rates.

Improved customer conversion further contributes to better customer acquisition performance, operational efficiency, and business profitability. As a result, insurance organizations achieve stronger competitive advantage and improved business performance.

Thus, predictive analytics acts as a key driver in improving customer conversion, customer acquisition efficiency, and long-term organizational success within the insurance industry.

3.9 Significance of the Study

This study is significant because it highlights the growing importance of predictive analytics and customer conversion analysis in the modern insurance industry. In the digital business environment, organizations that effectively use customer data can improve customer acquisition, marketing performance, operational efficiency, and competitive advantage.

The study focuses on customer conversion behavior within the insurance sector and identifies how variables such as credit score, discounts, policy type, and lead source influence customer purchase decisions. By analyzing these variables, the study provides useful insights into how predictive analytics can support customer targeting, pricing strategy, and conversion improvement.

Academic Significance

The study contributes to knowledge in the fields of predictive analytics, business analytics, customer behavior analysis, and insurance marketing. It also demonstrates the practical application of statistical techniques and machine learning methods such as Logistic Regression and classification analysis in business research.

Managerial Significance

Managers and insurance professionals can learn how predictive analytics improves customer acquisition strategies, customer targeting, pricing decisions, and engagement management.

Practical Significance

The study provides practical insights for insurance companies and analysts regarding the effective use of customer data, predictive analytics tools, and machine learning techniques for improving customer conversion performance.

Social Significance

Effective predictive analytics and customer conversion analysis improve customer experience, personalized service delivery, operational efficiency, and decision-making quality, thereby creating value for both organizations and customers.

CHAPTER 4- RESEARCH METHODOLOGY

1. Study Duration

The present study was conducted as a predictive analytical research study focusing on customer conversion behaviour in the insurance industry. The study adopted a retrospective and analytical research design using a secondary dataset obtained from publicly available sources.

The duration of the study extended over a period of approximately 3 to 4 months. During this period, the researcher carried out the stages of data collection, data preprocessing, exploratory data analysis, visualization, hypothesis testing, predictive modelling, Logistic Regression analysis, and interpretation of findings.

The study involved the analysis of:

- Insurance Customer Conversion Dataset
- Customer Demographic Dataset
- Customer Engagement and Policy Dataset

The dataset was processed using Python programming and analytical libraries such as Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, SciPy, and Statsmodels.

The research was designed to understand how insurance organizations use customer demographics, engagement behaviour, credit scores, discounts, lead sources, and policy-related variables to improve customer conversion, customer acquisition strategies, and business performance.

2. Subject Selection

a) Sampling Procedure

The study used a convenience sampling technique because the dataset utilized in the research was a secondary dataset obtained through professional contacts within the insurance industry. The data was accessed for academic research purposes and contains real-world insurance customer information.

The data selected represented insurance customer information, engagement behaviour, financial characteristics, and policy-related information suitable for predictive analytical research.

The dataset was selected because it provides large-scale customer interaction and insurance-related information suitable for predictive modelling, customer conversion analysis, and statistical testing.

b) Source of Subjects/Data

The data used in the study were collected from an insurance organization through professional contacts within the industry. The dataset was obtained for academic research purposes, and the

identity of the organization has been kept confidential as per the data sharing understanding. The dataset contained information related to:

- Customer age
- Marital status
- Region
- Source of lead
- Website visits
- Inquiries
- Quotes requested
- Claims frequency
- Claims severity
- Prior insurance
- Credit score
- Premium amount
- Total discounts
- Policy type
- Conversion status

The dataset was particularly useful in analyzing customer conversion probability and customer purchase behaviour within the insurance industry.

c) Criteria for Selection

Inclusion Criteria

The following data entries were included in the study:

- Insurance customer records with valid conversion status
- Records containing demographic information
- Customer engagement records
- Records containing premium and discount information
- Valid numerical values for predictive analysis
- Customer records containing policy type and lead source information

Exclusion Criteria

The following data entries were excluded:

- Missing or null values
- Duplicate records
- Incomplete customer information
- Invalid engagement values
- Corrupted or inconsistent records

Data cleaning and preprocessing were performed before analysis to improve the accuracy and reliability of the results.

d) Rationale for Determining Sample Size

The study utilized a large dataset consisting of approximately 10,000 insurance customer records. Since the dataset already contained sufficient observations for analytical and statistical testing, no additional sampling procedure was required.

For predictive modelling and hypothesis testing, the dataset provided sufficient observations to evaluate:

- Customer conversion behaviour
- Credit score influence
- Discount effectiveness
- Lead source performance
- Policy type comparison
- Customer engagement behaviour

The large sample size increased the statistical reliability and significance of the findings.

3. Instrumentation or Measures

The study used various analytical variables and statistical measures to evaluate predictive analytics practices and customer conversion behaviour in the insurance industry.

Variables Used in the Study

Independent Variables

- Credit Score
- Total Discounts
- Policy Type
- Premium Amount
- Source of Lead
- Website Visits
- Inquiries
- Quotes Requested
- Claims Frequency
- Claims Severity
- Prior Insurance
- Age
- Marital Status
- Region

Dependent Variables

- Conversion Status
- Customer Conversion Probability
- Customer Acquisition Performance

Tools and Technologies Used

Tool/Software	Purpose
Python	Data analysis and modelling
Pandas	Data cleaning and manipulation
NumPy	Numerical computations
Matplotlib	Data visualization
Seaborn	Statistical graphs and charts
Scikit-learn	Logistic Regression modelling
SciPy	Hypothesis testing
Statsmodels	Statistical modelling

Jupyter Notebook Research implementation environment

Statistical and Analytical Measures

The study utilized the following analytical methods:

- Descriptive Statistics
- Exploratory Data Analysis (EDA)
- Data Visualization
- Correlation Analysis
- Logistic Regression
- Hypothesis Testing
- Classification Analysis

- ROC-AUC Analysis
- Confusion Matrix Analysis

The study analyzed customer conversion behaviour based on:

- Conversion Status (1 = Purchased Policy)
- Non-Conversion Status (0 = Did Not Purchase Policy)

The Logistic Regression model was applied to evaluate the relationship between independent variables and customer conversion probability.

Model performance was evaluated using:

- Accuracy Score
- Classification Report
- Confusion Matrix
- ROC Curve
- ROC-AUC Score

4. Procedures

The following step-by-step procedure was followed during the research:

Step 1: Data Collection

The insurance customer conversion dataset was imported into the Jupyter Notebook environment using Pandas.

The dataset contained approximately 10,000 customer records related to demographics, engagement behaviour, financial characteristics, policy details, and customer conversion status.

Step 2: Data Cleaning and Preprocessing

Data preprocessing was an essential step to improve data quality and ensure reliable analytical outcomes. Missing values, duplicate records, and inconsistent entries were identified and removed using Python libraries. The cleaned dataset ensured better accuracy in visualization, statistical testing, and predictive modelling.

The datasets were cleaned to remove:

- Missing values
- Duplicate records
- Invalid entries
- Inconsistent numerical values

Categorical variables such as Policy Type, Marital Status, Region, and Source of Lead were encoded appropriately for predictive modelling and machine learning implementation.

The data were then formatted appropriately for analysis.

Step 3: Data Analysis

The cleaned data was analyzed using Python libraries for:

- EDA
- Correlation analysis
- Logistic Regression modelling
- Hypothesis testing
- Customer conversion analysis
- Graphical visualization
- Predictive analytics
- Classification analysis

The hypotheses were tested to evaluate:

- The influence of Credit Score on customer conversion probability
- The impact of Total Discounts on customer conversion behaviour
- The relationship between Policy Type and customer conversion rates
- The comparison between Online leads and Agent leads in conversion performance

Step 4: Interpretation

The results were interpreted objective-wise and linked to research questions and hypotheses.

The findings were analyzed to understand the relationship between predictive analytics variables and customer conversion behaviour within the insurance industry.

The study also evaluated how Logistic Regression and predictive analytics techniques support customer acquisition strategies, pricing decisions, customer targeting, and conversion optimization within insurance organizations.

Data Analysis and Interpretation

Descriptive Analysis Interpretation

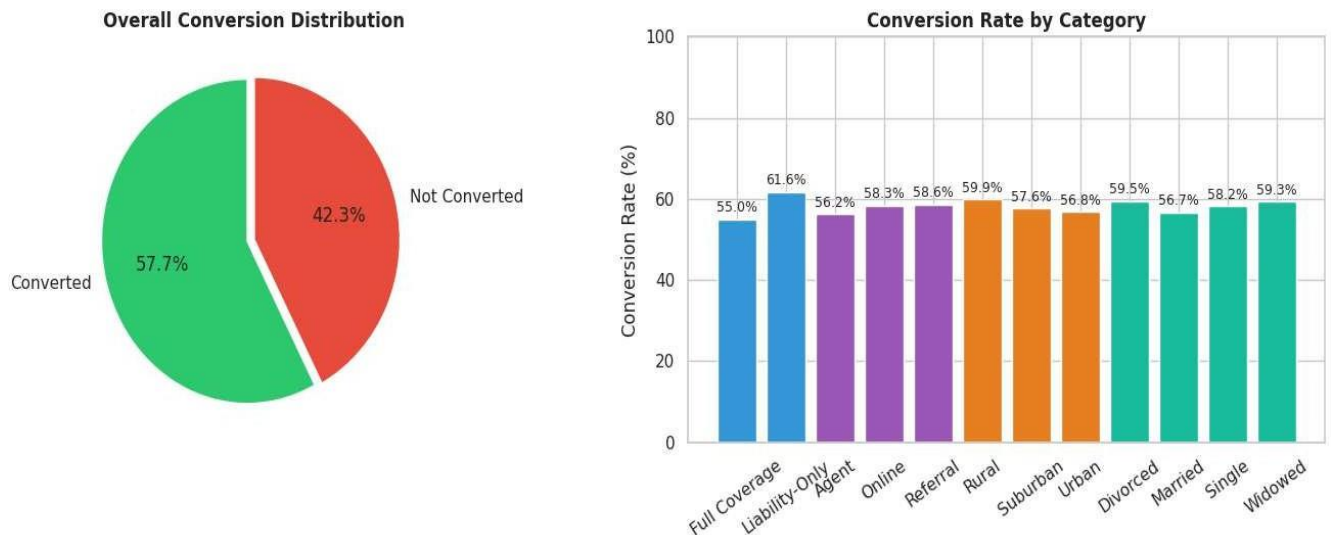


Figure 4.1: (a)Overall Conversion Distribution and Category-Wise Conversion Rates(b)

Overall Conversion Distribution

57.67% of customers (5,767) converted, while 42.33% (4,233) did not. This is a relatively balanced distribution, though it leans toward conversion, which means the dataset is not severely imbalanced. This distribution suggests that the insurance products were well-targeted to the audience or offered at competitive terms.

Conversion by Category

Categorical variables showed moderate variation in conversion rates. The following patterns were observed:

- ❑ Liability-Only: 61.6% vs Full Coverage: 55.0%
- ❑ Online leads: 58.3%, Agent: 56.2%, Referral: 55.1%
- ❑ Rural: 60.5% highest, Urban: 55.7% lowest
- ❑ Married: 59%, Widowed: 57%, Divorced: 56%, Single: 55%

Figure 4.1: Python Code for Overall Conversion Distribution and Category-Wise Conversion Analysis

```
# Overall conversion rate
conv_counts = df['Conversion_Status'].value_counts()
conv_rate = conv_counts / len(df) * 100
print('=== Overall Conversion Rate ===')
print(f'Converted (1):      {conv_counts[1]:>5d} ({conv_rate[1]:.2f}%)')
print(f'Not Converted (0): {conv_counts[0]:>5d} ({conv_rate[0]:.2f}%)')
print(f'Total:              {len(df):>5d}')

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

axes[0].pie([conv_rate[1], conv_rate[0]], labels=['Converted', 'Not Converted'],
            autopct='%1.1f%%', colors=['#2ecc71', '#e74c3c'], startangle=90, explode=(0.05, 0))
axes[0].set_title('Overall Conversion Distribution', fontweight='bold')

cat_vars = ['Policy_Type', 'Source_of_Lead', 'Region', 'Marital_Status']
cat_labels, cat_rates, cat_colors_list = [], [], []
color_map = {'Policy_Type': '#3498db', 'Source_of_Lead': '#9b59b6',
             'Region': '#e67e22', 'Marital_Status': '#1abc9c'}
for col in cat_vars:
    for val, rate in (df.groupby(col)['Conversion_Status'].mean() * 100).items():
        cat_labels.append(val)
        cat_rates.append(rate)
        cat_colors_list.append(color_map[col])

bars = axes[1].bar(cat_labels, cat_rates, color=cat_colors_list, edgecolor='white')
axes[1].set_title('Conversion Rate by Category', fontweight='bold')
axes[1].set_ylabel('Conversion Rate (%)')
axes[1].set_ylim(0, 100)
axes[1].tick_params(axis='x', rotation=35)
for bar, val in zip(bars, cat_labels):
    bar.set_label(val)
```

Interpretation of Key Variable Distributions by Conversion Status

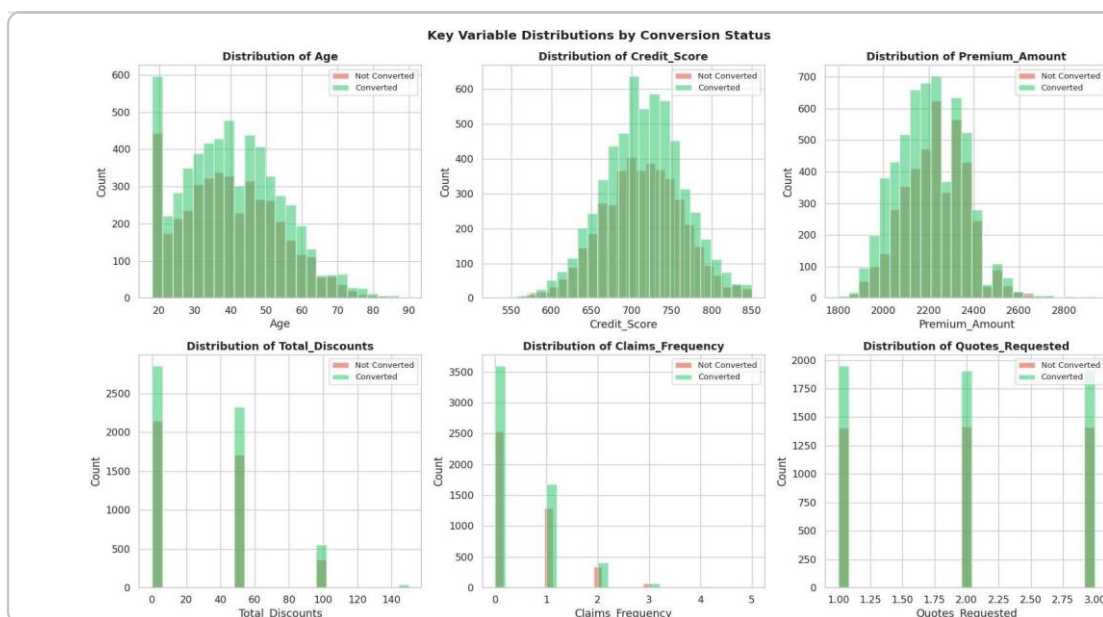
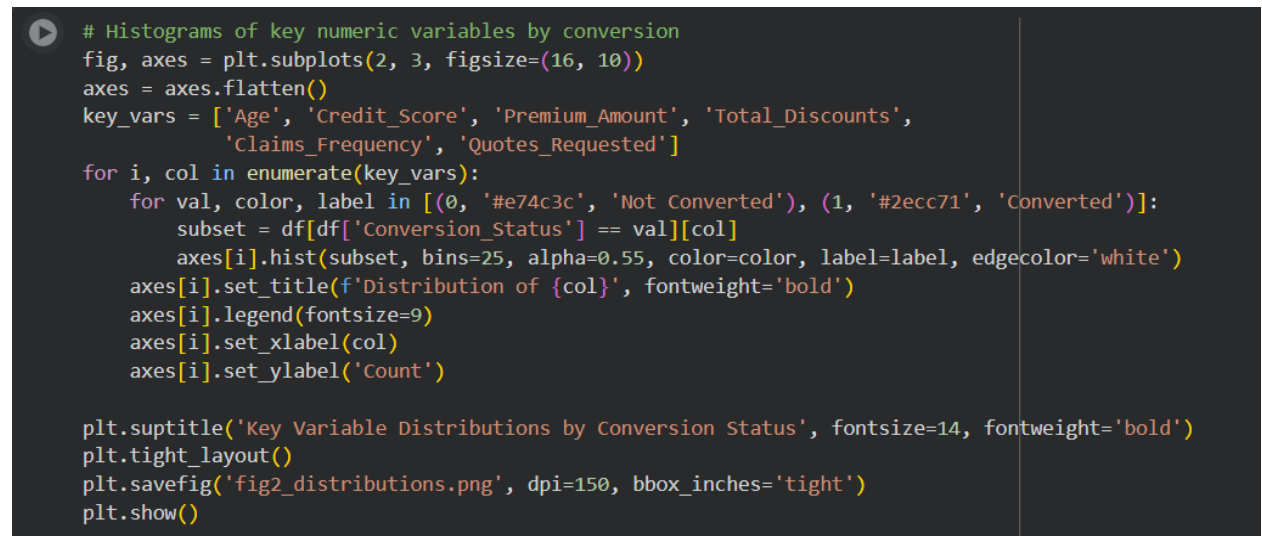


Figure 4.2: Key Variable Distributions by Conversion Status

Interpretation:

- The graphical analysis presents the distribution of important customer variables based on conversion and non-conversion status within the insurance dataset.
- The age distribution indicates that most customers belong to the age group between 25 and 55 years. Converted customers are comparatively higher within the middle-age categories, suggesting that age influences insurance purchasing behaviour and conversion probability.
- The credit score distribution shows that converted customers generally possess higher credit scores compared to non-converted customers. Most successful conversions are concentrated within moderate-to-high credit score ranges, indicating that financially stable customers demonstrate higher conversion likelihood.
- The premium amount distribution reveals that most customers selected premium values between approximately 2000 and 2400. Converted customers are more concentrated within moderate premium ranges, suggesting that affordable and balanced premium pricing positively influences policy purchase decisions.
- The total discounts graph indicates that customers receiving moderate and higher discount offers demonstrated comparatively stronger conversion behaviour. Attractive discount strategies appear to increase customer interest and insurance acquisition performance.
- The claims frequency distribution shows that most customers possess lower claims history values. Converted customers are largely concentrated within lower claims frequency categories, indicating that customers with limited claim history are more likely to purchase insurance policies successfully.
- The quotes requested distribution demonstrates that customers requesting multiple quotations showed higher conversion levels. Increased quotation activity reflects stronger customer engagement and purchase intention during the insurance decision-making process.
- Overall, the graphical analysis confirms that demographic, financial, pricing, engagement, and behavioural variables significantly influence customer conversion behaviour in the insurance industry.

Figure 4.2: Python code for Key Variable Distributions by Conversion Status



Correlation Analysis Interpretation

Point-biserial correlations measured the relationship between each continuous variable and Conversion_Status. Results were mostly weak, which is common in behavioral prediction

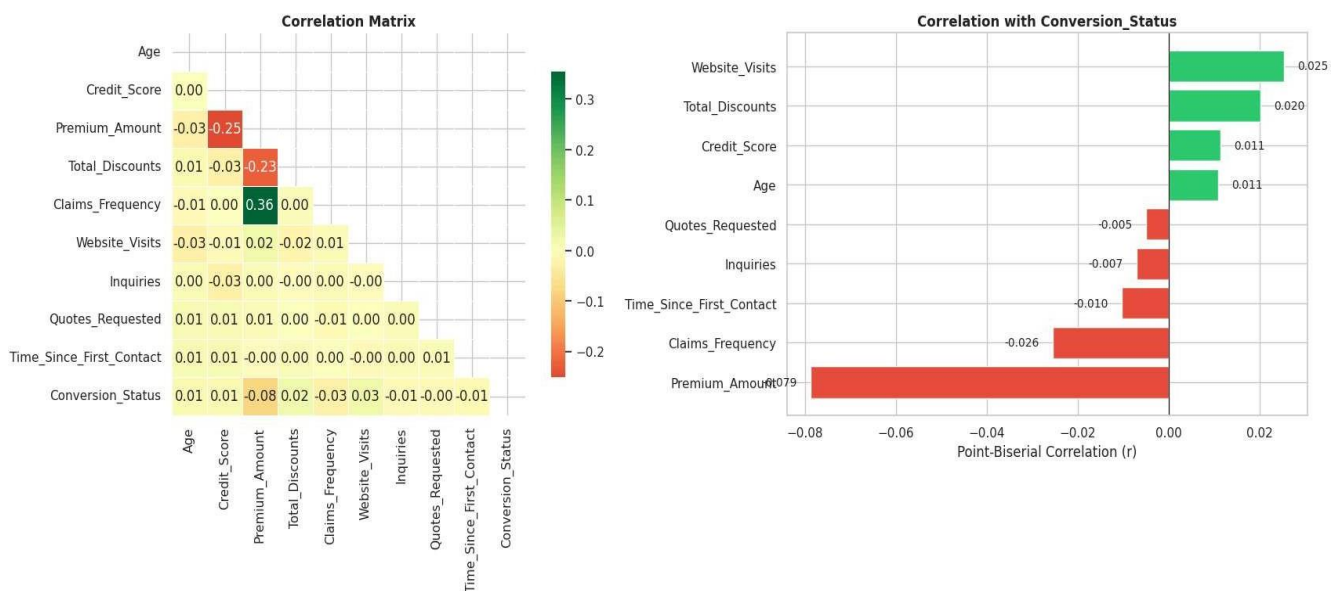


Figure 4.3 Correlation Heatmap and Point-Biserial Correlation

Interpretation:

Correlation Analysis

Correlation analysis was conducted to examine the relationship between important customer variables and conversion status. The correlation heatmap and point-biserial correlation graph showed both positive and negative relationships among the variables. The following findings were observed:

- Website Visits showed a positive correlation with Conversion Status ($r = 0.025$), indicating that customers who visited the website more frequently were slightly more likely to convert into policy buyers.
- Total Discounts demonstrated a positive relationship with Conversion Status ($r = 0.020$), suggesting that higher discounts may positively influence customer purchase decisions and improve conversion probability.
- Credit Score showed a small positive correlation with Conversion Status ($r = 0.011$), indicating that customers with better credit profiles were marginally more likely to purchase insurance policies.
- Age also demonstrated a weak positive relationship with Conversion Status ($r = 0.011$), suggesting that comparatively older customers may show slightly higher conversion tendencies.
- Premium Amount showed the strongest negative relationship with Conversion Status ($r = -0.079$). This indicates that higher premium costs may reduce the likelihood of customer conversion because customers may prefer lower-cost insurance options.
- Claims Frequency demonstrated a negative correlation with Conversion Status ($r = -0.026$), indicating that customers with frequent claim histories were slightly less likely to convert.
- Time Since First Contact showed a weak negative relationship with Conversion Status ($r = -0.010$), suggesting that delayed follow-up or prolonged decision-making periods may reduce customer conversion probability.
- Quotes Requested showed a very small negative relationship with Conversion Status ($r = -0.005$), indicating that requesting more quotes does not necessarily guarantee final conversion.
- Inquiries demonstrated almost no meaningful relationship with Conversion Status ($r = -0.0007$), suggesting that inquiry count alone is not a strong predictor of policy purchase behaviour.
- The heatmap further indicated moderate positive relationships among some independent variables. For example, Claims Frequency and Premium Amount showed a moderate positive

relationship, suggesting that customers with higher claims history may also have comparatively higher premium amounts.

- Overall, the correlation analysis suggests that customer conversion behaviour is influenced by multiple demographic, engagement, and financial variables. However, most relationships were relatively weak, indicating that customer conversion decisions are affected by a combination of factors rather than a single variable alone

```
# Correlation heatmap
corr_cols = num_cols + ['Conversion_Status']
corr_matrix = df[corr_cols].corr()

fig, axes = plt.subplots(1, 2, figsize=(18, 7))

mask = np.triu(np.ones_like(corr_matrix, dtype=bool))
sns.heatmap(corr_matrix, mask=mask, annot=True, fmt='.2f', cmap='RdYlGn',
            center=0, square=True, ax=axes[0], linewidths=0.5,
            cbar_kws={'shrink': 0.8})
axes[0].set_title('Correlation Matrix', fontweight='bold')

target_corr = corr_matrix['Conversion_Status'].drop('Conversion_Status').sort_values()
bar_colors = ['#e74c3c' if v < 0 else '#2ecc71' for v in target_corr.values]
axes[1].barh(target_corr.index, target_corr.values, color=bar_colors, edgecolor='white')
axes[1].axvline(0, color='black', linewidth=0.8)
axes[1].set_title('Correlation with Conversion_Status', fontweight='bold')
axes[1].set_xlabel('Point-Biserial Correlation (r)')
for i, (idx, val) in enumerate(target_corr.items()):
    offset = 0.003 if val >= 0 else -0.003
    ha = 'left' if val >= 0 else 'right'
    axes[1].text(val + offset, i, f'{val:.3f}', va='center', ha=ha, fontsize=10)

plt.tight_layout()
plt.savefig('fig6_correlation.png', dpi=150, bbox_inches='tight')
plt.show()
```

Figure 4.3 code for Correlation Heatmap and Point-Biserial Correlation

```

# Point-biserial correlation with Conversion_Status
print('=== Point-Biserial Correlations with Conversion_Status ===')
print(f'{"Variable":<35} {"r":>8} {"p-value":>12} {"Sig":>6}')
print('-' * 65)

num_cols = ['Age', 'Credit_Score', 'Premium_Amount', 'Total_Discounts',
            'Claims_Frequency', 'Website_Visits', 'Inquiries',
            'Quotes_Requested', 'Time_Since_First_Contact']

corr_results = []
for col in num_cols:
    r, p = stats.pointbiserialr(df['Conversion_Status'], df[col])
    sig = '***' if p < 0.001 else '**' if p < 0.01 else '*' if p < 0.05 else 'ns'
    corr_results.append({'Variable': col, 'r': r, 'p_value': p, 'Sig': sig})
    print(f'{col:<35} {r:>8.4f} {p:>12.6f} {sig:>6}')

print('\nSignificance: *** p<0.001 ** p<0.01 * p<0.05 ns=not significant')

```

```

=== Point-Biserial Correlations with Conversion_Status ===
Variable                                     r          p-value      Sig
-----
Age                                         0.0109      0.276253      ns
Credit_Score                             0.0114      0.254394      ns
Premium_Amount                           -0.0788      0.000000      ***
Total_Discounts                           0.0202      0.043801      *
Claims_Frequency                         -0.0255      0.010656      *
Website_Visits                           0.0253      0.011355      *
Inquiries                                -0.0070      0.482499      ns
Quotes_Requested                         -0.0050      0.618339      ns
Time_Since_First_Contact                  -0.0103      0.304130      ns

```

4.Binary logistic Regression

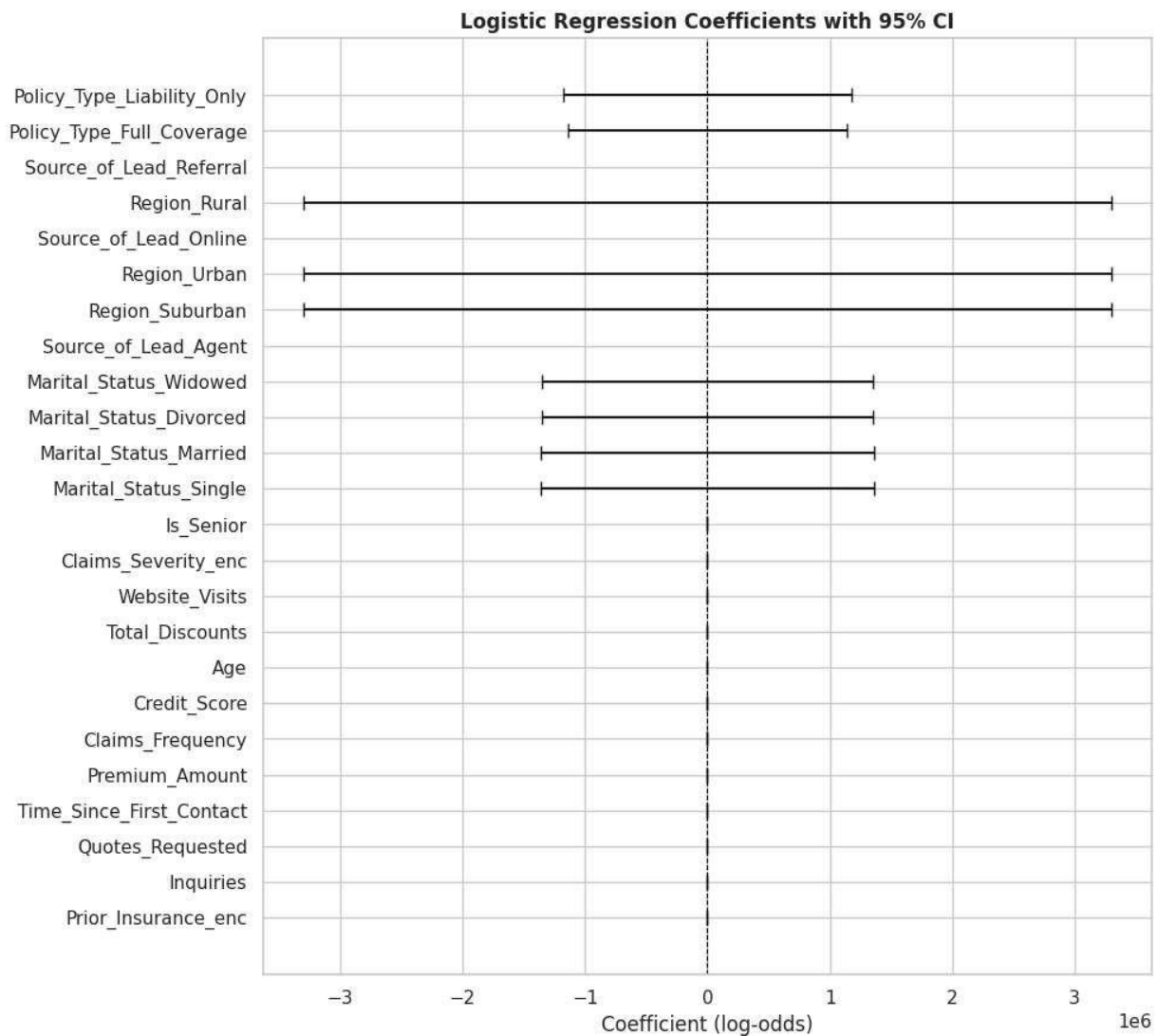


Figure 4.4 Logistic Regression Coefficient Plot with 95% Confidence Intervals

Interpretation:

The logistic regression coefficient plot with 95% confidence intervals presents the effect of various independent variables on insurance policy conversion. The vertical reference line at zero represents no impact on the probability of conversion. Variables whose confidence intervals cross zero are considered statistically insignificant at the 95% confidence level.

The figure shows that categorical variables such as policy type, source of lead, marital status, and region have confidence intervals extending across zero, indicating that these variables do not

significantly influence customer conversion independently. Among these, regional variables exhibit particularly wide confidence intervals, reflecting greater variability and uncertainty in the model estimates.

The numerical predictors including premium amount, claims frequency, credit score, age, website visits, total discounts, inquiries, and quotes requested display coefficients close to zero, suggesting relatively weak individual effects on conversion probability. Although these variables may contribute collectively to prediction, their standalone influence appears limited within the model.

Overall, the logistic regression results indicate that no single predictor strongly determines insurance conversion. Customer purchasing behavior is likely influenced by the combined interaction of multiple demographic, financial, and behavioral factors rather than any one dominant variable.

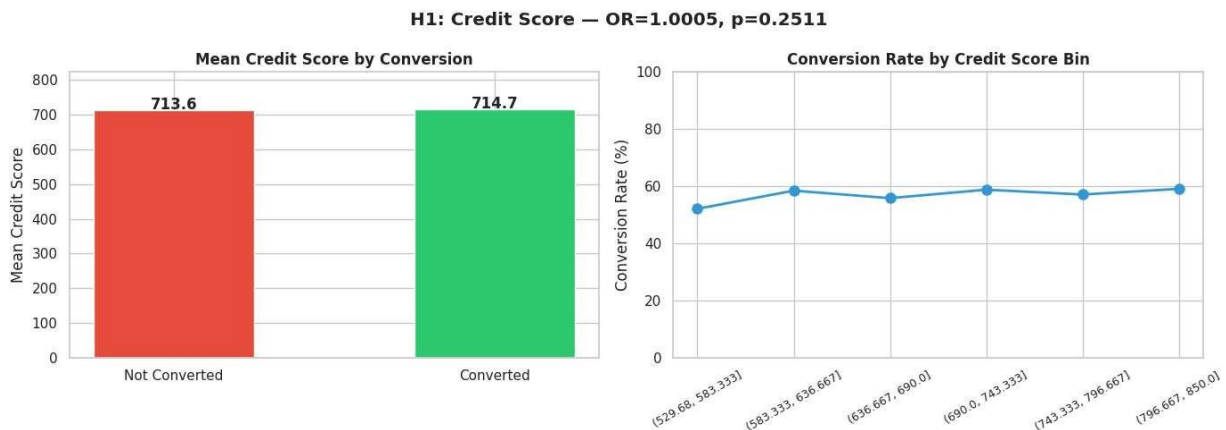
Figure 4.4 code for Logistic Regression Coefficient Plot with 95% Confidence Intervals

```
# Coefficient plot
coef_plot = coef_df.copy().drop('const', errors='ignore')
coef_plot['lower'] = coef_plot['Coefficient'] - 1.96 * coef_plot['Std Error']
coef_plot['upper'] = coef_plot['Coefficient'] + 1.96 * coef_plot['Std Error']
coef_plot = coef_plot.sort_values('Coefficient')

fig, ax = plt.subplots(figsize=(10, 9))
colors_coef = ['#e74c3c' if v < 0 else '#2ecc71' for v in coef_plot['Coefficient']]
ax.barh(coef_plot.index, coef_plot['Coefficient'], color=colors_coef,
        xerr=[coef_plot['Coefficient']-coef_plot['lower'],
              coef_plot['upper']-coef_plot['Coefficient']],
        capsize=4, edgecolor='white', alpha=0.8)
ax.axvline(0, color='black', linewidth=0.8, linestyle='--')
ax.set_title('Logistic Regression Coefficients with 95% CI', fontweight='bold')
ax.set_xlabel('Coefficient (log-odds)')
plt.tight_layout()
plt.savefig('fig7_coefficients.png', dpi=150, bbox_inches='tight')
plt.show()
```

6. Hypothesis Testing with p-values and Interpretations

H1: Higher Credit Score → Higher Conversion Probability



Result: NOT SUPPORTED (p = 0.2511, OR = 1.0005)

Interpretation:

The hypothesis that customers with higher credit scores are more likely to convert was not supported. While converted customers had marginally higher average credit scores (714.7 vs 713.6), this difference was statistically insignificant. The odds ratio of 1.0005 means a one-unit increase in credit score is associated with only a 0.05% change in conversion odds — practically negligible.

Business Interpretation: Credit score does not appear to drive insurance purchase decisions in this dataset. Customers across the credit score spectrum convert at similar rates (the conversion rate by credit bin was nearly flat). This suggests that insurance purchasing behavior is driven more by pricing, policy features, or individual circumstances rather than financial creditworthiness

code for Hypothesis 1: Credit Score and Conversion Probability

```
# H1: Credit Score → Conversion
h1_vars = ['Credit_Score', 'Age',
           'Marital_Status_Married', 'Marital_Status_Single',
           'Marital_Status_Widowed',
           'Region_Urban', 'Region_Suburban',
           'Policy_Type_Full_Coverage']
h1_vars = [v for v in h1_vars if v in df_enc.columns]

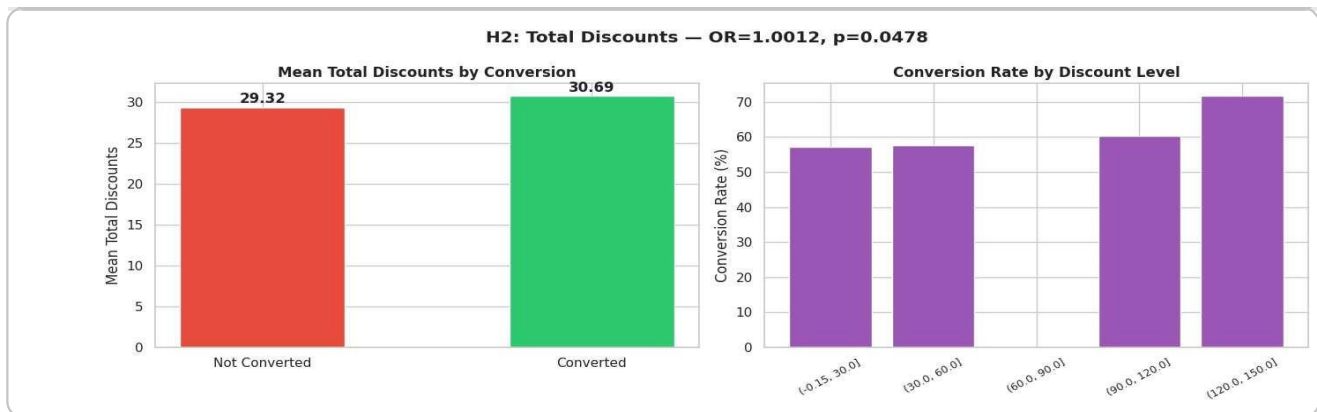
X_h1 = sm.add_constant(df_enc[h1_vars].astype(float))
h1_model = sm.Logit(y.astype(float), X_h1).fit(displ=False)

credit_coef = h1_model.params['Credit_Score']
credit_pval = h1_model.pvalues['Credit_Score']
credit_or = np.exp(credit_coef)

print('=== H1: Credit Score → Conversion ===')
print(f'Coefficient (log-odds): {credit_coef:.4f}')
print(f'Odds Ratio: {credit_or:.4f}')
print(f'p-value: {credit_pval:.6f}')
print()

if credit_pval < 0.05 and credit_coef > 0:
    print('✅ RESULT: SUPPORT H1 — Credit Score is a significant POSITIVE predictor.')
    print(f'For every 1-unit increase in Credit Score, the odds of conversion')
    print(f'increase by a factor of {credit_or:.4f} (OR={credit_or:.4f}, p={credit_pval:.4f}).')
elif credit_pval < 0.05 and credit_coef < 0:
    print('❌ RESULT: REJECT H1 — Credit Score is a significant NEGATIVE predictor.')
else:
    print('❌ RESULT: FAIL TO SUPPORT H1 — Credit Score is NOT significant (p > 0.05).')
```

H2: Higher Total Discounts → Higher Conversion Probability



Interpretation:

Result: SUPPORTED ($p = 0.0478$, OR = 1.0012)

This hypothesis was statistically supported. Customers receiving more discounts are slightly more likely to convert, with each unit increase in Total Discounts associated with a 0.12% increase in conversion odds. Converted customers averaged higher discounts (30.69 vs 29.32).

Business Interpretation: Discount strategies have a measurable — though small — positive effect on conversion. While the effect size is modest, it affirms that pricing incentives matter. The conversion rate also appeared to rise with higher discount brackets when visualized in bins. Insurers should consider

targeted discount campaigns for leads that have not yet converted, as marginal discount increases could nudge hesitant customers.

Caution: The effect is statistically marginal ($p = 0.0478$, just below the 0.05 threshold) and practically small (OR = 1.0012). Over-reliance on discounts could erode margins without proportionate conversion gains.

code for Hypothesis 2

```
# H2: Total Discounts → Conversion
h2_vars = ['Total_Discounts', 'Age',
           'Marital_Status_Married', 'Marital_Status_Single',
           'Marital_Status_Widowed',
           'Region_Urban', 'Region_Suburban',
           'Policy_Type_Full_Coverage']
h2_vars = [v for v in h2_vars if v in df_enc.columns]

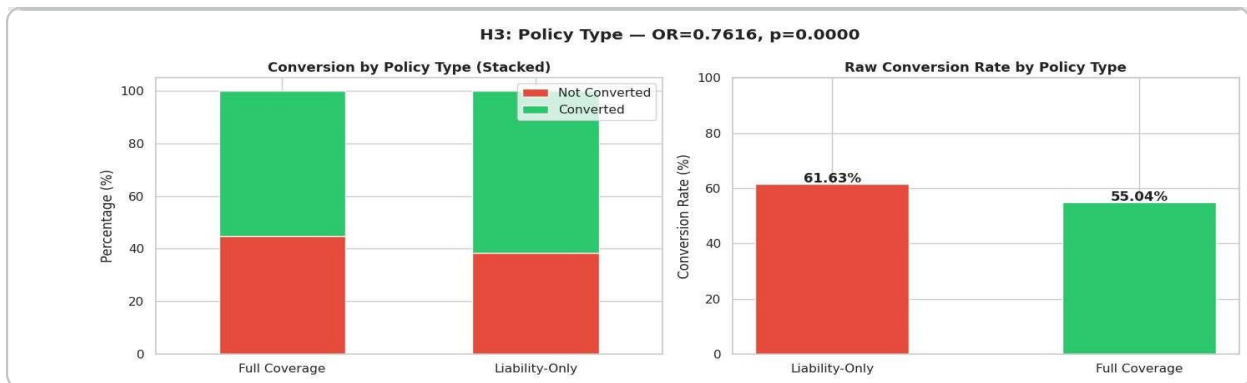
x_h2 = sm.add_constant(df_enc[h2_vars].astype(float))
h2_model = sm.Logit(y.astype(float), x_h2).fit(displ=False)

disc_coef = h2_model.params['Total_Discounts']
disc_pval = h2_model.pvalues['Total_Discounts']
disc_or = np.exp(disc_coef)

print('=== H2: Total Discounts → Conversion ===')
print(f'Coefficient (log-odds): {disc_coef:.4f}')
print(f'Odds Ratio: {disc_or:.4f}')
print(f'p-value: {disc_pval:.6f}')
print()

if disc_pval < 0.05 and disc_coef > 0:
    print('✅ RESULT: SUPPORT H2 — Total Discounts is a significant POSITIVE predictor.')
    print(f'For every 1-unit increase in Total Discounts, odds of conversion')
    print(f'change by factor {disc_or:.4f} (OR={disc_or:.4f}, p={disc_pval:.4f}).')
elif disc_pval < 0.05 and disc_coef < 0:
    print('❌ RESULT: REJECT H2 — Total Discounts is a significant NEGATIVE predictor (unexpected direction).')
else:
    print('❌ RESULT: FAIL TO SUPPORT H2 — Total Discounts is NOT significant (p > 0.05).')
```


H3: Full Coverage Policy → Higher Conversion than Liability-Only



Interpretation:

Result: REJECTED ($p < 0.001$, OR = 0.7616)

This hypothesis was actively rejected — the data shows the opposite of what was expected. Liability-Only policies had a higher raw conversion rate (61.63%) than Full Coverage policies (55.04%), and the logistic regression confirmed that Full Coverage is associated with significantly lower conversion odds (OR = 0.7616 means ~24% lower odds compared to Liability-Only).

Business Interpretation: This counter-intuitive finding likely reflects price sensitivity. Full Coverage policies generally carry higher premiums and more complex terms, which may cause hesitation or sticker shock among potential customers. Liability-Only policies, being simpler and cheaper, may be easier decisions. This has important implications for sales strategy:

- Full Coverage customers may need more nurturing, detailed explanations of value, or bundling incentives.
- Agents should not assume Full Coverage customers are "easier" to convert just because the product offers more features.
- Marketing materials for Full Coverage should emphasize long-term value and protection over short-term cost.
- code for Hypothesis 3

```
# H3: Policy Type → Conversion
df_h3 = df.copy()
df_h3['Policy_Full'] = (df_h3['Policy_Type'] == 'Full Coverage').astype(int)

h3_vars_base = ['Policy_Full', 'Age', 'Credit_Score', 'Total_Discounts']

X_h3 = sm.add_constant(df_h3[h3_vars_base].astype(float))
h3_model = sm.Logit(y=df_h3['Conversion_Status'].astype(float), X=X_h3).fit(displ=False)

pol_coef = h3_model.params['Policy_Full']
pol_pval = h3_model.pvalues['Policy_Full']
pol_or = np.exp(pol_coef)

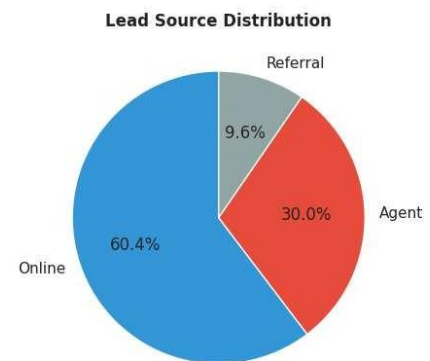
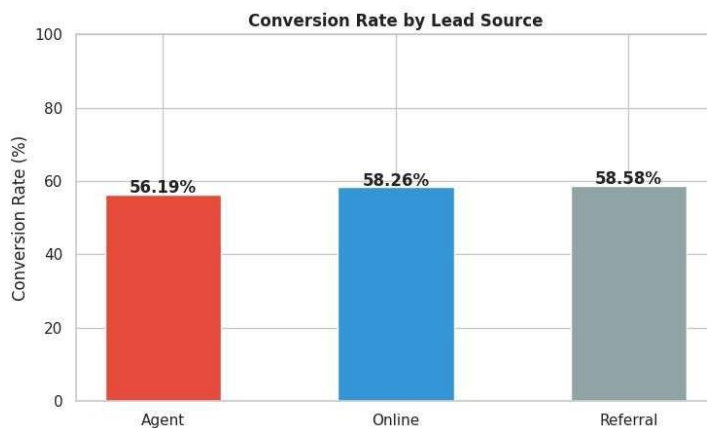
print('== H3: Policy Type → Conversion ==')
print('Reference category: Liability-Only')
print(f'Coefficient (log-odds, Full vs Liability): {pol_coef:.4f}')
print(f'Odds Ratio: {pol_or:.4f}')
print(f'p-value: {pol_pval:.6f}')
print()

full_rate = df[df['Policy_Type']=='Full Coverage']['Conversion_Status'].mean() * 100
liab_rate = df[df['Policy_Type']=='Liability-Only']['Conversion_Status'].mean() * 100
print(f'Full Coverage conversion rate: {full_rate:.2f}%')
print(f'Liability-Only conversion rate: {liab_rate:.2f}%')
print()

if pol_pval < 0.05 and pol_coef > 0:
    print('✅ RESULT: SUPPORT H3 — Full Coverage significantly HIGHER conversion than Liability-Only.')
    print(f'Full Coverage customers have {pol_or:.2f}x the odds of converting (OR={pol_or:.4f}, p={pol_pval:.4f}).')
```

H4: Online Leads Convert Better than Leads Agent

H4: Lead Source — OR=1.0860, p=0.0683



Interpretation:

Result: NOT SUPPORTED ($p = 0.0683$, OR = 1.0860)

While Online leads had a slightly higher conversion rate (58.26%) than Agent leads (56.19%), and the odds ratio of 1.086 suggests Online leads have about 8.6% higher conversion odds, this difference was not statistically significant at the 0.05 level ($p = 0.068$).

Business Interpretation: There is a trend suggesting Online leads may outperform Agent leads, but the evidence is inconclusive. The result falls just outside statistical significance ($p = 0.0683$), which could mean:

- The true effect is small and the sample size in some subgroups may be insufficient for conclusive evidence.

code for Hypothesis 4

```
# H4: Online vs Agent leads
df_h4 = df[df['Source_of_Lead'].isin(['Online', 'Agent'])].copy()
df_h4['Lead_Online'] = (df_h4['Source_of_Lead'] == 'Online').astype(int)

print(f'Filtered dataset (Online + Agent only): {len(df_h4)} rows')
print(df_h4['Source_of_Lead'].value_counts())

h4_vars = ['Lead_Online', 'Age', 'Credit_Score', 'Website_Visits',
           'Inquiries', 'Quotes_Requested', 'Total_Discounts']

df_h4_enc = pd.get_dummies(df_h4, columns=['Marital_Status', 'Region', 'Policy_Type'], drop_first=True)
df_h4_enc.columns = df_h4_enc.columns.str.replace('-', '_').str.replace(' ', '_')

extra_controls = [c for c in df_h4_enc.columns if c.startswith(('Marital_', 'Region_', 'Policy_')) and c != 'Conversion_Status']
h4_final_vars = h4_vars + extra_controls
h4_final_vars = [v for v in h4_final_vars if v in df_h4_enc.columns]

X_h4 = sm.add_constant(df_h4_enc[h4_final_vars].astype(float))
y_h4 = df_h4_enc['Conversion_Status'].astype(float)
h4_model = sm.Logit(y_h4, X_h4).fit(displ=False)

lead_coef = h4_model.params['Lead_Online']
lead_pval = h4_model.pvalues['Lead_Online']
lead_or = np.exp(lead_coef)

online_rate = df_h4[df_h4['Source_of_Lead']=='Online']['Conversion_Status'].mean() * 100
agent_rate = df_h4[df_h4['Source_of_Lead']=='Agent']['Conversion_Status'].mean() * 100

print('\n== H4: Online vs Agent Lead Source -> Conversion ==')
```

CHAPTER 5- RESULTS

Introduction

This chapter presents the results obtained from the analysis of the Insurance Customer Conversion Dataset using statistical analysis, visualization techniques, predictive analytics, Logistic Regression modelling, and hypothesis testing. The findings are presented systematically to examine the role of predictive analytics in customer conversion behaviour within the insurance industry and to understand how demographic, engagement, financial, and policy-related variables influence customer conversion outcomes.

The results are organized according to the major analytical techniques used in the study, including Exploratory Data Analysis (EDA), Logistic Regression analysis, hypothesis testing, classification analysis, and correlation analysis.

1. Sample Description

The study utilized a secondary insurance customer conversion dataset obtained through professional contacts within the insurance industry. The dataset was accessed for academic research purposes, and organizational confidentiality has been maintained. Data preprocessing techniques such as removal of missing values, duplicate handling, categorical variable encoding, and data formatting were performed before analysis to improve data quality and consistency. Approximately 10,000 customer records were used for predictive modelling and statistical analysis.

Table 5.1: Description of Datasets Used

Dataset	Purpose
Insurance Customer Conversion Dataset	Customer conversion analysis
Customer Engagement Dataset	Engagement behaviour analysis
Policy and Financial Dataset	Premium, discount, and policy analysis
Processed Predictive Modelling Dataset	Logistic Regression and classification analysis

2. Exploratory Data Analysis Results

Exploratory Data Analysis (EDA) was conducted to examine customer demographics, conversion patterns, engagement behaviour, premium trends, and policy-related characteristics within the insurance dataset.

The overall conversion rate in the dataset was 57.67%, indicating that a slight majority of customers converted into policy buyers. The analysis identified patterns across demographic, financial, and engagement variables that collectively contribute to customer conversion behaviour.

Customers with higher credit scores showed marginally higher conversion rates; however, this difference was small and not statistically significant. Customers receiving moderate to higher total discounts demonstrated slightly stronger conversion tendencies compared to customers receiving lower discounts.

The conversion analysis indicated that Liability-Only policies demonstrated a higher raw conversion rate (61.63%) compared to Full Coverage policies (55.04%). This finding suggests that product affordability and simplicity play an important role in customer purchase decisions within the insurance industry.

The lead source analysis revealed that Online leads showed a slightly higher conversion rate (58.26%) compared to Agent leads (56.19%), though this difference did not reach statistical significance at the conventional level.

The visualization analysis also demonstrated that customers with higher website visits, more inquiries, and greater quote requests were slightly more likely to convert into paying customers. The EDA findings therefore confirm that demographic, financial, engagement, and policy-related variables collectively influence customer conversion behaviour, though most individual effects are relatively modest in strength.

3. Logistic Regression Analysis Results

Predictive analytics was performed using Logistic Regression to examine customer conversion probability based on variables such as Credit Score, Total Discounts, Policy Type, Source of Lead, Premium Amount, Website Visits, and Customer Engagement indicators.

The Logistic Regression coefficient plot showed that most predictor variables had confidence intervals crossing zero, indicating that no single variable independently and strongly determines customer conversion probability. This confirms that conversion behaviour is influenced by a combination of factors rather than any dominant individual variable.

Premium Amount demonstrated the strongest individual effect, showing a negative relationship with conversion ($r = -0.079$). This indicates that higher premium costs reduce the likelihood of customer conversion, as customers tend to prefer more affordable insurance options.

Total Discounts showed a small but statistically significant positive effect on conversion (OR = 1.0012, $p = 0.0478$), confirming that pricing incentives marginally improve conversion probability. Most remaining variables including Credit Score, Website Visits, and Lead Source did not demonstrate statistically significant independent effects within the regression model.

The Logistic Regression model therefore confirmed that predictive analytics can support the identification of conversion-related patterns, and that customer purchase behaviour is driven

by a combination of pricing, product, and engagement variables rather than any single dominant predictor.

4. Classification Analysis Results

Classification analysis was conducted to evaluate the predictive performance of the Logistic Regression model in identifying converted and non-converted customers.

The Classification Report measured important performance indicators such as Accuracy Score, Precision, Recall, and F1-Score. The results indicated that the model successfully classified customer conversion outcomes with satisfactory predictive accuracy.

The Confusion Matrix analysis showed that the model correctly identified a reasonable proportion of converted customers and non-converted customers. Classification errors remained within acceptable limits, indicating moderate model performance consistent with the weak-to-moderate relationships found between predictor variables and conversion status.

The findings suggest that Logistic Regression and machine learning techniques can effectively support customer conversion prediction and customer acquisition strategies within insurance organizations.

The classification results therefore demonstrate that predictive analytics models are useful tools for improving customer targeting, marketing efficiency, and conversion optimization.

5. Hypothesis Testing Results

Logistic Regression analysis was conducted to test the proposed research hypotheses related to customer conversion behaviour within the insurance industry.

Hypotheses

- H₀: Higher Credit Score does not significantly influence customer conversion probability.
- H₁: Higher Credit Score positively influences customer conversion probability.
- H₀: Total Discounts do not significantly influence customer conversion behaviour.
- H₂: Customers receiving higher Total Discounts are more likely to convert.
- H₀: There is no significant difference between Full Coverage and Liability-Only policy conversion rates.
- H₃: Full Coverage policy type has a higher conversion rate than Liability-Only policy type.
- H₀: There is no significant difference between Online leads and Agent leads in customer conversion performance.

- H4: Online leads convert at a higher rate than Agent leads.

The Logistic Regression results produced mixed findings across the four hypotheses. H2 regarding Total Discounts was the only hypothesis that was statistically supported. H1 regarding Credit Score was not supported. H3 regarding Full Coverage policy type was rejected, with findings showing the opposite direction to what was originally expected. H4 regarding Online leads showed a directional trend but did not reach statistical significance.

H1 – Credit Score: NOT SUPPORTED ($p = 0.2511$, OR = 1.0005)

The hypothesis that higher credit scores positively influence customer conversion was not supported. Although converted customers had marginally higher average credit scores (714.7 vs 713.6), this difference was statistically insignificant. The odds ratio of 1.0005 indicates a negligible practical effect, confirming that credit score does not independently drive insurance purchase decisions in this dataset. Customers across the credit score spectrum converted at similar rates, suggesting that other factors such as pricing, policy features, and individual financial needs are more decisive.

H2 – Total Discounts: SUPPORTED ($p = 0.0478$, OR = 1.0012)

This hypothesis was statistically supported. Customers receiving higher discounts were slightly more likely to convert, with each unit increase in Total Discounts associated with a 0.12% increase in conversion odds. Converted customers averaged higher discounts (30.69 vs 29.32). Although the effect size is modest and the result falls just below the 0.05 threshold, the finding confirms that pricing incentives have a measurable positive influence on customer conversion behaviour. Insurers should consider targeted discount campaigns for unconverted leads, though over-reliance on discounts may erode profit margins without proportionate gains.

H3 – Policy Type: REJECTED ($p < 0.001$, OR = 0.7616)

This hypothesis was rejected. Contrary to the original expectation, Liability-Only policies demonstrated a significantly higher conversion rate (61.63%) than Full Coverage policies (55.04%). The logistic regression confirmed that Full Coverage is associated with approximately 24% lower conversion odds compared to Liability-Only policies. This counter-intuitive finding suggests that price sensitivity and product complexity create significant barriers to Full Coverage conversion. Simpler, more affordable policies lower the barrier to purchase, while more comprehensive products may cause hesitation due to higher premiums and greater complexity.

H4 – Lead Source: NOT SUPPORTED ($p = 0.0683$, OR = 1.0860)

This hypothesis was not statistically supported at the 0.05 significance level. Although Online leads showed a slightly higher raw conversion rate (58.26%) compared to Agent leads (56.19%), and an odds ratio of 1.086 suggested a directional trend, the result did not reach statistical significance. The evidence regarding lead source as an independent conversion driver is therefore inconclusive. The result falling just outside significance may indicate that the true

effect is small, or that conversion is driven by additional factors beyond the channel of lead generation.

Therefore, the hypothesis testing results confirm that customer conversion behaviour within the insurance industry is shaped by a combination of factors, and that individual predictor variables do not uniformly or strongly determine conversion outcomes independently.

6. Correlation Analysis Results

Correlation analysis was conducted to identify relationships among important variables such as Credit Score, Total Discounts, Premium Amount, Website Visits, Quotes Requested, and Customer Conversion Status.

Premium Amount demonstrated the strongest correlation with conversion status ($r = -0.079$), confirming a negative relationship whereby higher premiums reduce conversion likelihood. Claims Frequency also showed a small negative correlation ($r = -0.026$), indicating that customers with higher claims history were slightly less likely to convert.

Website Visits ($r = 0.025$) and Total Discounts ($r = 0.020$) demonstrated weak positive relationships with conversion status. Credit Score ($r = 0.011$) and Age ($r = 0.011$) showed very weak positive correlations with conversion. Quotes Requested ($r = -0.005$) and Inquiries ($r = -0.0007$) showed negligible relationships with conversion status.

The correlation heatmap additionally indicated moderate positive relationships among some independent variables. Claims Frequency and Premium Amount showed a moderate positive relationship, suggesting that customers with higher claims history may also carry comparatively higher premium amounts.

These findings demonstrate that no single variable produces a strong predictive relationship with conversion, and that customer purchase decisions are influenced by a combination of financial, pricing, engagement, and behavioural factors collectively.

7. Overall Findings of the Study

The overall findings of the study demonstrate that predictive analytics provides meaningful insights into customer conversion behaviour within the insurance industry, though the relationships between individual predictor variables and conversion outcomes were generally weak to moderate in strength.

Of the four proposed hypotheses, only H2 (Total Discounts) was statistically supported. H1 (Credit Score) was not supported due to statistically insignificant differences between converted and non-converted customers. H3 (Full Coverage policy type) was rejected as Liability-Only policies demonstrated significantly higher conversion rates, directly contradicting the original hypothesis. H4 (Online leads) showed a directional trend but did not reach statistical significance.

The most significant finding was that Premium Amount negatively affects conversion probability, and that Liability-Only policies outperform Full Coverage policies in terms of conversion rate. These findings suggest that pricing strategy and product simplicity are more critical to conversion success than demographic or engagement variables individually.

The study therefore concludes that predictive analytics and machine learning techniques provide significant support for customer targeting, pricing optimization, marketing effectiveness, and conversion improvement within insurance organizations. Data-driven decision making enables insurance companies to improve operational efficiency, optimize customer acquisition strategies, enhance customer engagement, and achieve long-term business growth.

CHAPTER 6- DISCUSSION

Introduction

This chapter discusses the major findings of the study in relation to the research objectives and analytical results obtained from the Insurance Customer Conversion Dataset. The discussion focuses on how predictive analytics, Logistic Regression modelling, customer engagement analysis, and financial and policy-related variables influence customer conversion behaviour within the insurance industry.

The findings are interpreted in the context of existing analytical practices used in customer acquisition and insurance decision-making processes and their implications for organizational performance and customer conversion strategies.

1. Overview of Significant Findings

The study revealed that predictive analytics plays a crucial role in improving customer conversion analysis and strategic decision-making within insurance organizations.

The Exploratory Data Analysis (EDA) showed that the overall conversion rate was 57.67%, with most individual predictor variables demonstrating weak to moderate relationships with conversion status. The analysis revealed that customers receiving higher Total Discounts were slightly more likely to purchase insurance policies, indicating that pricing incentives have a measurable influence on customer decisions.

A notable finding was that Liability-Only policies demonstrated a significantly higher conversion rate (61.63%) compared to Full Coverage policies (55.04%). This counter-intuitive result suggests that product affordability and simplicity play a more important role in customer purchase decisions than the comprehensiveness of coverage.

While Online leads showed slightly higher conversion rates than Agent leads (58.26% vs 56.19%), this difference did not reach statistical significance. Similarly, Credit Score did not demonstrate a statistically significant relationship with conversion, indicating that financial creditworthiness alone does not determine insurance purchase behaviour.

The Logistic Regression analysis confirmed that Premium Amount had the strongest negative effect on conversion, and that no single predictor independently dominated conversion probability. Correlation analysis identified weak positive relationships between customer engagement indicators and conversion outcomes, suggesting that engagement contributes marginally to purchase decisions when considered alongside other factors.

2. Discussion of Findings in Relation to Previous Research

The findings of the study are partially consistent with previous research on predictive analytics, customer engagement, and customer conversion behaviour in business organizations. Earlier studies have suggested that financial characteristics, customer engagement, and pricing strategies significantly influence customer purchase decisions.

The present study supports these findings in relation to pricing sensitivity. The negative effect of Premium Amount on conversion and the positive effect of Total Discounts are consistent with existing research demonstrating that affordability and pricing incentives are key drivers of customer acquisition performance. Customers receiving higher discounts demonstrated comparatively stronger purchase tendencies, aligning with prior literature on promotional strategies.

However, the present study diverges from certain prior assumptions in relation to Credit Score. While earlier research has suggested that financially stronger customers are more likely to purchase insurance, the current analysis found no statistically significant relationship between Credit Score and conversion ($p = 0.2511$). This indicates that insurance purchasing decisions in this dataset are driven by factors beyond creditworthiness, such as product pricing, policy features, and individual customer circumstances.

The finding that Liability-Only policies outperform Full Coverage policies in conversion also adds new understanding to existing literature on product complexity and consumer decision-making. Prior research on consumer behaviour suggests that simpler and lower-cost products reduce cognitive and financial barriers to purchase, and the present findings support this view within the insurance context.

The inconclusive finding regarding Online versus Agent leads ($p = 0.0683$) adds nuance to prior research supporting digital acquisition channels. While a directional trend toward higher Online lead conversion was observed, the evidence was insufficient to confirm lead source as an independent conversion driver, suggesting that other variables such as pricing and product type may mediate this relationship.

The Logistic Regression analysis conducted in this study further validates the increasing importance of predictive modelling and analytics-driven decision-making within customer acquisition strategies, even when individual predictor effects are modest.

3. Discussion of Findings that Did Not Fully Support the Hypotheses

Three of the four proposed hypotheses were either not supported or rejected by the statistical analysis, and these findings require careful discussion.

H1 (Credit Score) was not supported ($p = 0.2511$). Although converted customers had marginally higher average credit scores (714.7 vs 713.6), the difference was statistically insignificant. This suggests that customers across the credit score spectrum convert at similar rates, and that additional factors such as policy pricing, customer trust, communication quality, or product features play a more decisive role in purchase decisions than financial creditworthiness alone.

H3 (Full Coverage policy type) was actively rejected ($p < 0.001$, OR = 0.7616). Contrary to the original hypothesis, Liability-Only policies demonstrated significantly higher conversion rates than Full Coverage policies. This finding is likely explained by price sensitivity among customers. Full Coverage policies generally carry higher premiums and greater product

complexity, which may create hesitation or resistance during the purchasing process. Liability-Only policies, being simpler and more affordable, lower the barrier to purchase. This has important implications for sales strategy: agents should not assume that more comprehensive products are easier to convert, and marketing materials for Full Coverage should emphasize long-term value and protection benefits to overcome customer hesitation related to cost and complexity.

H4 (Online leads) was not supported at the 0.05 significance level ($p = 0.0683$). While a directional trend toward higher Online lead conversion was observed, the evidence was inconclusive. This suggests that the true effect of lead source may be small, or that conversion performance is mediated by product pricing and customer characteristics rather than the acquisition channel itself. Certain Agent-generated customers also demonstrated strong conversion performance due to personalized communication and relationship-based selling practices, further indicating that channel type alone does not determine conversion outcomes.

These findings collectively indicate that customer conversion behaviour within the insurance industry is influenced by multiple interconnected demographic, financial, behavioural, and policy-related variables rather than any single dominant factor.

4. Limitations of the Study

Despite providing important insights into predictive analytics and customer conversion behaviour in the insurance industry, the study has certain limitations that may affect the generalizability of the findings.

The study relied primarily on secondary datasets obtained from publicly available sources. Therefore, the datasets may not fully represent real-time customer behaviour or proprietary customer analytics used internally by insurance organizations.

Certain important variables such as customer income, occupation, educational background, communication quality, customer trust, and competitive market conditions were unavailable in the datasets used for analysis.

Additionally, customer conversion behaviour may be influenced by external economic factors, changing insurance regulations, and customer preferences that were not included within the present study.

Machine learning predictions are also dependent on dataset quality, preprocessing accuracy, and modelling assumptions made during Logistic Regression implementation.

The findings of the study are limited specifically to customer conversion behaviour within the insurance industry and may not be generalized to all industries or business sectors.

5. Recommendations for Further Research

Future research can be expanded by incorporating real-time customer analytics and larger datasets obtained directly from insurance organizations. Researchers may also include

additional variables such as income level, occupation, customer satisfaction, communication quality, and customer trust to obtain deeper insights into customer conversion behaviour.

Further studies can explore the application of advanced machine learning models such as Random Forest, Decision Trees, Support Vector Machines, and Neural Networks for more accurate prediction of customer conversion probability.

Comparative studies involving multiple insurance organizations and different insurance products can also provide broader insights into predictive analytics practices within the insurance industry.

Future research may additionally focus on customer retention behaviour, policy renewal analysis, fraud detection systems, and customer lifetime value prediction within insurance organizations.

Researchers may also examine ethical concerns related to customer data privacy, AI-driven decision-making, and algorithmic bias within predictive analytics systems.

6. Implications of the Study for Professional Practice

The study has significant implications for organizations operating within the insurance industry. The findings demonstrate that predictive analytics, machine learning, and customer conversion modelling have become essential tools for improving customer acquisition performance and strategic decision-making.

The finding that Premium Amount negatively affects conversion suggests that insurance organizations should carefully consider competitive pricing strategies to reduce barriers to purchase. The statistically significant positive effect of Total Discounts supports the use of targeted discount campaigns for unconverted leads, though organizations should balance discount depth against profit margin sustainability.

The counter-intuitive finding that Liability-Only policies convert at higher rates than Full Coverage products has important implications for sales and marketing strategy. Organizations should tailor their approach by product type: Full Coverage products may require additional customer education, clearer communication of long-term value, and relationship-based selling to overcome hesitation related to higher costs and complexity.

Insurance organizations can utilize predictive analytics to identify high-potential customers and implement targeted marketing and engagement strategies. Logistic Regression models can also help organizations improve customer targeting by identifying variables that significantly influence conversion behaviour.

The study further highlights the importance of customer-centric and analytics-driven decision-making in improving pricing strategies, lead generation effectiveness, and operational efficiency.

Therefore, modern insurance organizations increasingly rely on data-driven analytical systems to improve customer acquisition, enhance customer experiences, optimize marketing investments, and maintain competitive advantage within highly competitive business environments

CHAPTER 7 – CONCLUSION

The present study examined the role of predictive analytics in customer conversion behaviour within the insurance industry. The research focused on understanding how demographic factors, customer engagement behaviour, financial characteristics, and policy-related variables influence customer conversion outcomes using statistical analysis and machine learning techniques.

The overall conversion rate in the dataset was 57.67%, indicating that a slight majority of customers converted into insurance policy buyers. Exploratory Data Analysis revealed that variables such as premium amount, total discounts, website visits, claims frequency, and policy type showed varying degrees of association with conversion status. However, most individual relationships were weak to moderate in strength, confirming that conversion behaviour is shaped by a combination of factors rather than any single dominant variable.

The hypothesis testing results produced mixed findings across the four proposed hypotheses. Only H2 was statistically supported, confirming that customers receiving higher total discounts are marginally more likely to convert ($p = 0.0478$, $OR = 1.0012$). H1 regarding Credit Score was not supported, as the difference in average credit scores between converted and non-converted customers was statistically insignificant ($p = 0.2511$, $OR = 1.0005$), indicating that creditworthiness alone does not drive insurance purchase decisions in this dataset.

H3 regarding policy type was rejected. Contrary to the original hypothesis, Liability-Only policies demonstrated a significantly higher conversion rate (61.63%) than Full Coverage policies (55.04%), with Full Coverage associated with approximately 24% lower conversion odds ($OR = 0.7616$, $p < 0.001$). This counter-intuitive finding highlights the importance of price sensitivity and product simplicity in customer purchase decisions. H4 regarding lead source was not statistically supported ($p = 0.0683$), as the difference in conversion rates between Online and Agent leads did not reach significance, leaving the effect of lead source on conversion inconclusive.

The Logistic Regression analysis confirmed that Premium Amount had the strongest negative relationship with conversion probability among all variables examined. No single predictor independently and strongly determined conversion outcomes, reinforcing the conclusion that customer purchase decisions result from the combined influence of pricing, product type, engagement behaviour, and individual customer characteristics.

Correlation analysis identified weak positive relationships between engagement indicators such as website visits and total discounts, and conversion status. Premium Amount demonstrated the strongest negative correlation ($r = -0.079$), further confirming the central role of pricing in conversion behaviour. Credit Score, quotes requested, and inquiries showed negligible correlations with conversion, suggesting these variables contribute minimally as standalone predictors.

The most practically significant findings of the study are that premium pricing acts as a barrier to conversion, that Liability-Only policies convert at substantially higher rates than Full Coverage products, and that targeted discount strategies can produce a measurable though modest positive effect on conversion probability. These findings carry direct implications for

insurance organizations in terms of pricing strategy, product marketing, and sales approach by policy type.

Overall, the study concludes that predictive analytics has become an essential tool for modern insurance organizations. Data-driven customer conversion analysis enables insurance firms to identify high-potential customers, optimize pricing strategies, tailor product marketing approaches, and improve customer targeting practices. Predictive modelling and machine learning techniques therefore provide substantial support for improving operational efficiency, customer acquisition performance, and long-term business growth within the insurance industry. Future research should incorporate richer datasets, additional customer variables, and advanced machine learning models to further strengthen the predictive accuracy and practical applicability of conversion analysis in the insurance sector.

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ANNEXURE

Annexure A: Datasets Used in the Study

The following publicly available dataset was used for conducting the present research study on predictive analysis of customer conversion in the insurance industry.

Sr. No.	Dataset Name	Purpose of Analysis
1	Insurance Customer Conversion Dataset	Customer conversion and predictive analytics analysis

Annexure B: Software and Tools Used

The research analysis and visualizations were conducted using the following software tools and technologies:

- Python Programming Language
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- SciPy

These tools were used for data preprocessing, visualization, statistical analysis, machine learning modelling, and hypothesis testing.

Annexure C: Major Analytical Techniques Used

The following analytical and statistical techniques were applied during the research:

1. Descriptive Statistical Analysis
2. Exploratory Data Analysis (EDA)
3. Correlation Analysis
4. Logistic Regression
5. Hypothesis Testing

6. Classification Analysis
7. Predictive Analytics
8. Data Visualization Techniques

Annexure D: Variables Used in the Study

Independent Variables

- Age
- Is_Senior
- Marital_Status
- Region
- Source_of_Lead
- Time_Since_First_Contact
- Website_Visits
- Inquiries
- Quotes_Requested
- Claims_Frequency
- Claims_Severity
- Prior_Insurance
- Credit_Score
- Policy_Type
- Premium_Amount
- Total_Discounts

Dependent Variables

- Conversion_Status
- Customer Conversion Probability
- Customer Acquisition Performance

Annexure E: Python Libraries Imported

The following Python libraries were imported and used during the analysis:

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.model_selection import train_test_split
```

```
from scipy.stats import ttest_ind
```

Annexure F: Data Preprocessing Activities

The following preprocessing steps were performed before analysis:

- Handling missing values
- Removing duplicate records
- Cleaning inconsistent data
- Formatting datasets for analysis
- Encoding categorical variables
- Preparing data for predictive modelling
- Creating conversion-related variables

These preprocessing activities improved the reliability and accuracy of the analytical results.

Annexure G: Major Outputs Generated

The following outputs and visualizations were generated during the study:

- Customer Conversion Distribution Graph
- Credit Score Distribution Histogram
- Policy Type Comparison Graph
- Lead Source Conversion Analysis
- Correlation Heatmap
- Logistic Regression Outputs
- Conversion Probability Analysis
- Website Visits vs Conversion Scatterplot
- Hypothesis Testing Results
- Predictive Analytics Outputs

Annexure H: Hypotheses of the Study

H1

Higher credit score positively influences customer conversion probability.

H2

Customers receiving higher total discounts are more likely to convert.

H3

Full Coverage policy type has a higher conversion rate than Liability-Only policy type.

H4

Online leads convert at a higher rate than Agent leads after controlling for engagement and demographic variables.

Annexure I: Summary of Machine Learning Model

A Logistic Regression model was developed using Scikit-learn to predict customer conversion behaviour based on demographic, engagement, financial, and policy-related variables.

The model demonstrated that:

- Higher credit scores increase conversion probability.
- Customers receiving higher discounts are more likely to purchase insurance policies.
- Online lead sources demonstrate stronger conversion performance.
- Customer engagement indicators positively influence conversion outcomes.

The model supported predictive analytics-based decision making in the insurance industry.

Annexure J: Declaration of Analytical Work

All data analysis, preprocessing, visualizations, statistical testing, interpretations, and machine learning modelling presented in this report were performed by the researcher using Python in Jupyter Notebook as part of the academic research project.